

Mechanical and electrical engineers

80 practical engineering questions for evidence-aware building-services investigation, testing, maintenance, commissioning and scenario assessment.

STAKEHOLDER	16
QUESTION RANGE	ME-001 to ME-080
COVERAGE	Eight engineering categories, ten questions each
EVIDENCE STANCE	Proposed sensing + authoritative records + explicit safe failure

ENGINEERING USE BOUNDARY

This catalogue supports engineering enquiry. Sensors, integrations and analytics are proposed until independently verified. Current authoritative owners and competent professionals take precedence. Talking Abacws does not provide engineering sign-off, approve designs, operate or isolate systems, certify compliance or safety, or replace inspection, testing, commissioning and competent-person judgement.

Evidence before inference

Proposed-until-verified rule

Every sensor, integration and analytic remains proposed until deployment, exact physical mapping, commissioning, permission and current health are independently verified. Each record checks modality/location, spatial and temporal coverage, cadence/aggregation, calibration or field verification, clock alignment, uptime, gaps, drift and provenance.

Historical spatial reference

The May 2021 floor plan is historical spatial reference only. It cannot establish current room use, access, services, hazards, closures, plant condition or safe work areas. Present-tense claims require current authoritative records and competent site evidence.

Authority and safe failure

Controlled issue/effective time, physical observation/test time, telemetry observed time and retrieval time remain distinct. Current authoritative owners, duty holders, responsible persons and competent professionals take precedence. Conflicts stay visible. Missing or weak minimum evidence returns NOT VERIFIED and the next safe evidence step.

Engineering boundary

Talking Abacws is not a design authority, authorised person, duty holder, permit issuer, commissioning authority or statutory certifier. It does not switch, isolate, reset, override or alter control logic. Telemetry does not replace prove-safe procedures, electrical tests, airflow/flow/pressure tests, balancing, water-hygiene controls or competent engineering judgement.

Six fields, one engineering decision

1. Stakeholder question

A natural, role-specific question framed around an engineering decision, test, survey, maintenance action or scenario.

2. Sensors and telemetry

Proposed modality and location, coverage, cadence/aggregation, measurement limitations and the per-record quality gate.

3. Authoritative supporting data

Current source owners and precedence, issue/effective/test/observed/retrieved times, permissions, provenance, staleness and conflicts.

4. Analysis required

Operation type, decision interval, minimum evidence, method, uncertainty and a safe failure state when evidence is insufficient.

5. Answer boundary

The specific inference, disclosure, promise, certification, authorisation or control action that the service must not make.

6. Why this is useful

The concrete next decision: survey, test, correction, maintenance, recommissioning, design study, escalation or scenario selection.

EVIDENCE CLOCKS

Issue/effective time says when a controlled record applies. Observation/test time says when physical evidence occurred. Telemetry observed time says when a point was sampled. Retrieved time says when Talking Abacws obtained it. Retrieval never makes old evidence current.

State the smallest claim the evidence supports

SUPPORTED

Current required sources are mapped, permissioned and sufficiently consistent for the bounded statement.

INDICATIVE

A proxy, incomplete coverage, model assumption or material uncertainty prevents a direct engineering conclusion.

CONFLICTED

Current records, competent field/test evidence or telemetry disagree; the disagreement remains visible until resolved.

NOT VERIFIED

Minimum evidence is absent, stale, unhealthy, unpermissioned or physically unverified; return the next safe evidence step.

CATALOGUE MAP

1	Drawings, asset registers and design intent	ME-001 to ME-010
2	Electrical distribution, loads and resilience	ME-011 to ME-020
3	HVAC plant, distribution and environmental performance	ME-021 to ME-030
4	Water, drainage and public-health systems	ME-031 to ME-040
5	Controls, BMS, metering and system integration	ME-041 to ME-050
6	Diagnostics, testing, commissioning and evidence quality	ME-051 to ME-060
7	Maintenance, statutory duties and technical risk	ME-061 to ME-070
8	Retrofit, capacity, lifecycle, decarbonisation and scenarios	ME-071 to ME-080

Minimum-evidence principle: keep lookup, observation, calculation, estimate, forecast, comparison, diagnosis and recommendation distinct. Missing evidence is not permission to infer. Association, timing, proxy and model output are not presented as causal proof or direct measurement.

Drawings, asset registers and design intent

ME-001 Where does the current controlled electrical single-line diagram conflict with the distribution physically evidenced today?

SENSORS AND TELEMETRY

Proposed: Read-only incomer/switchboard/DB current and voltage, breaker auxiliary status where fitted, verified meter-to-board mappings, and authorised field observations of panel labels. Coverage: Every board and feeder represented in the candidate SLD; unmetered or inaccessible branches are explicitly marked unknown. Timing: Native breaker events at 1-10 s where available; 15 min electrical summaries; compare event windows around known switching undertaken by authorised staff. Limits: Breaker status is not cable topology; current flow cannot reveal spare ways, hidden interconnections or protective settings. CT/VT ratios, phase assignment, units and board identity must be proven. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis. Decision: Before design, isolation planning, fault investigation or after a material electrical alteration. Minimum evidence: One current controlled SLD, current board schedules/identifiers, and independent physical or test evidence for every disputed node. Method: Match board IDs, feeder labels, device hierarchy and observed energisation; classify each link confirmed, conflicted, historical-only or unknown; issue a field-verification list. Uncertainty: Telemetry can corroborate energisation but cannot prove concealed cable routes, conductor size, spare capacity or protection coordination. Safe failure: Return NOT VERIFIED for any branch lacking current controlled and physical evidence, and require authorised site verification before engineering use.

WHY THIS IS USEFUL

The engineer can correct the information baseline and target only the panels/feeders that require field verification before design, shutdown or fault work.

AUTHORITATIVE SUPPORTING DATA

Owners: Estates electrical information owner, appointed electrical duty holder/responsible person, and authorised maintenance/test record owners. Precedence: Current controlled as-built SLD and panel schedules, current asset labels and approved change-control records, then competent-person inspection/test evidence; verified telemetry may corroborate but not define topology. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: A superseded drawing, unresolved field alteration, stale panel schedule or unhealthy/mis-mapped point makes the affected branch not verified. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not declare a diagram "as-built" from telemetry, infer concealed circuits, state that an isolation point is safe, or reveal restricted switchboard/security detail beyond the user's authorised role. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-002 Which mechanical plant and distribution paths are verified to serve this room or zone, and which links remain unverified?

SENSORS AND TELEMETRY

Proposed: Read-only AHU/fan status, damper/VAV feedback, valve position, supply/return temperatures, pump state and zone environmental points where those points are physically mapped. Coverage: The nominated zone plus candidate upstream AHUs, terminal units, heating/cooling branches and pumps shown by current schematics. Timing: 10-60 s plant states with 5 min summaries across several known operating cycles; use authorised functional events only as corroboration. Limits: Correlated movement does not prove service dependency; shared ducts, cross-connections, local units and undocumented alterations can create false associations. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis. Decision: Before plant shutdowns, zone alterations, complaint diagnosis or change-of-use assessment. Minimum evidence: A current schematic/zoning source plus at least one verified physical mapping from the zone to each claimed terminal or upstream system. Method: Trace upstream dependencies from room/zone to terminal, branch and plant; use synchronized states as supporting evidence; label each edge confirmed, probable, conflicted or unknown. Uncertainty: Telemetry cannot reveal concealed duct/pipe routing and shared systems may respond simultaneously without direct dependency. Safe failure: Return the last verified upstream node and a precise field-check requirement rather than extending the graph by inference.

WHY THIS IS USEFUL

The engineer can define shutdown impact, investigation scope and survey points without relying on an assumed service map.

AUTHORITATIVE SUPPORTING DATA

Owners: Estates mechanical information owner, BMS owner/operator, asset-information owner and current space-information owner. Precedence: Current controlled mechanical schematics/zoning drawings and verified asset/space mappings, then approved commissioning/balancing evidence and competent-person field findings; mapped BMS telemetry is corroborative. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: If the current room identity, asset tag or zone-to-terminal mapping is unresolved, the dependency remains conflicted regardless of telemetry correlation. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not infer current room use from the May 2021 floor plan, claim concealed routes, or state that upstream valves/dampers can be isolated safely. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-003 Which M&E asset-register records must be reconciled before they can support design, maintenance or lifecycle decisions?**SENSORS AND TELEMETRY**

Proposed: Read-only BMS point inventory, meter inventory, runtime/status presence and authorised field capture of physical asset tags or QR identifiers. Coverage: Selected electrical, HVAC, water and controls asset classes within the engineering scope, including non-instrumented assets when known from records. Timing: Current inventory snapshot plus 30 days of point freshness where useful; aggregate mapped/unmapped, duplicate and orphan counts by asset class. Limits: A point is not necessarily an asset; dormant assets may have no telemetry; aliases, reused tags and virtual points can create duplicate or false records. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + forecast. Decision: At project mobilisation, before maintenance migration and before lifecycle/capital modelling. Minimum evidence: Defined asset classes, current register extract, current drawing/CMMS identifiers and independent evidence for each contested asset. Method: Normalize asset ID, location, manufacturer/model, parent system and service; identify duplicate, orphan, missing-parent and unmapped records; assign evidence confidence per field. Uncertainty: Instrumented assets are easier to detect, so completeness scores are biased unless uninstrumented populations are included. Safe failure: Do not invent missing assets or merge ambiguous identifiers; return an evidence-gap list for owner resolution and field survey.

WHY THIS IS USEFUL

The engineer can decide which asset records are safe to use, which need correction, and where survey effort should be focused before design or maintenance planning.

AUTHORITATIVE SUPPORTING DATA

Owners: Estates asset-information owner, CMMS/maintenance owner, discipline engineers and BMS/metering information owners. Precedence: Current controlled asset register and CMMS master records, current drawings and approved handover/change records, then competent-person field identification; telemetry metadata is supporting evidence only. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Unknown tag provenance, conflicting serial/model data, retired-but-live records, or records with no current owner are reported as data defects rather than merged automatically. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not create, delete or overwrite authoritative asset records automatically, infer ownership from a BMS label, or present the analytical inventory as the legal/maintenance master. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-004 What duty was this plant designed and commissioned for, and what duty is it demonstrably delivering now?**SENSORS AND TELEMETRY**

Proposed: Asset-appropriate read-only quantities such as flow, differential pressure, temperatures, airflow, fan/pump speed and electrical input, plus valve/damper command/feedback where fitted. Coverage: The nominated plant item, its measurement stations and the minimum upstream/downstream context needed to calculate delivered duty. Timing: 1-5 min samples across representative load periods; retain raw/native values and calculate 15 min steady-state windows separately from transients. Limits: Derived thermal/air duties require valid sensor location, calibration and synchronized flow/temperature data; proxy speed or valve position is not delivered duty. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis + recommendation. Decision: During performance investigation, recommissioning, capacity assessment or before replacement sizing. Minimum evidence: Current design/selection duty, current configuration, verified measurement chain and at least three stable operating windows spanning meaningful load. Method: Calculate delivered duty with propagated sensor uncertainty; compare with design/commissioned operating envelope at comparable conditions; separate control limitation from physical capacity where possible. Uncertainty: Part-load conditions, fouling, weather and distribution constraints can make a plant appear undersized or oversized without proving root cause. Safe failure: Return a bounded duty range and the missing test/measurement needed; do not extrapolate to full design load from weak part-load data.

WHY THIS IS USEFUL

The engineer can decide whether to perform targeted testing, recommissioning, hydraulic/air balancing or a full capacity study before design changes.

AUTHORITATIVE SUPPORTING DATA

Owners: Discipline engineer, commissioning-record owner, asset-information owner and BMS/metering owner. Precedence: Current approved design duty/selection data and current commissioning/recommissioning test results, then current asset configuration and competent-person measurements; verified telemetry can show operational duty only within its measurement validity. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: If plant has been modified, design duty is superseded, test points are stale, or sensors lack calibration/location evidence, comparisons are indicative only. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not declare plant compliant with design, guarantee available capacity, infer internal condition solely from duty deviation, or size replacement plant from telemetry alone. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-005 Which current environmental design criteria govern this space, and which measured conditions can validly be compared with them?**SENSORS AND TELEMETRY**

Proposed: Zone temperature, relative humidity, CO2 and illuminance only where such sensors are mapped and suitable for the criterion; supply-air conditions may provide plant context. Coverage: The current engineering zone and the relevant environmental measurement locations; no individual workstation or person-level tracking. Timing: 1-5 min readings summarized into occupied-design windows only when the current schedule/use is authoritatively known; report percentile and duration rather than single excursions. Limits: Sensor location, shielding, drift and representativeness matter; CO2 is not a direct ventilation-rate measurement and comfort is not inferred from one sensor. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis. Decision: Before complaint investigation, change-of-use design, recommissioning or environmental setpoint assessment. Minimum evidence: A current authoritative criterion with its required measurement basis plus verified sensor location/accuracy and a representative operating window. Method: First test whether the available measurement is comparable to the criterion; then report duration/percentile outside the criterion with sensor uncertainty and operating context. Uncertainty: A zone sensor cannot establish conditions everywhere, and user experience may differ from measured environmental criteria. Safe failure: If the criterion, space use or measurement basis is not current, state NOT VERIFIED and request the appropriate brief or competent-person survey.

WHY THIS IS USEFUL

The engineer can compare measured conditions only against criteria that actually apply, then decide whether further environmental testing or design assessment is justified.

AUTHORITATIVE SUPPORTING DATA

Owners: Current space-information owner, discipline/design engineer, Estates operational owner and approved project/brief owner. Precedence: Current approved room/space brief and design criteria, current design documents and later approved change records, then competent-person survey/test results; historical floor-plan labels and generic defaults do not establish present requirements. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: If use/occupancy category or design criteria changed after the controlled document, the old criterion is historical-only until the owner confirms it. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not infer current room use, occupant satisfaction, health, productivity or legal compliance from environmental telemetry; do not treat the May 2021 floor plan as current design authority. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-006 Which isolation points and downstream dependencies are documented, and what must still be proved on site before work?**SENSORS AND TELEMETRY**

Proposed: Read-only switch/breaker auxiliary status, valve position feedback, pump/fan state and pressure/temperature decay only as contextual evidence after authorised operations. Coverage: The nominated asset/branch, documented isolation devices and dependent loads/services within the work boundary. Timing: Event-resolution data during an authorised isolation performed by competent staff; retain pre-event and post-event context, never initiate the action. Limits: Indication is not proof of isolation; auxiliary contacts may not prove contact separation, valves may pass, stored energy may remain, and undocumented cross-feeds can exist. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + calculation + diagnosis. Decision: During work planning and immediately before competent-person site verification. Minimum evidence: Current procedure/schematic, current asset labels, identified isolation devices and documented downstream/upstream dependencies. Method: Trace documented energy/service paths, flag alternative feeds and stored-energy sources, and produce a verification checklist; never calculate a "safe to work" state. Uncertainty: Records can omit temporary supplies, bypasses, local generation, backfeeds, pressure reservoirs or valve leakage. Safe failure: Return NOT VERIFIED where any path is unresolved and require the applicable lock-off, prove-dead, depressurisation or other competent-person procedure.

WHY THIS IS USEFUL

The engineer can prepare a focused isolation verification plan and identify dependency gaps before competent staff begin work.

AUTHORITATIVE SUPPORTING DATA

Owners: Electrical/mechanical duty holder or responsible person, permit/isolation procedure owner, discipline engineer and maintenance record owner. Precedence: Current approved isolation procedure/schematic and asset labels, then competent-person site identification and required proving/testing; telemetry and historical O&M information are context only. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Any unlabeled device, conflicting schematic, unresolved cross-feed, changed plant configuration or missing proving record makes the isolation path not verified. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not tell anyone that equipment is isolated, safe to touch or safe to open; do not instruct switching/valve operation or replace lockout/tagout, proving-dead or permit procedures. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-007 Are asset names and tags consistent across drawings, BMS, meters, O&M information and maintenance systems?**SENSORS AND TELEMETRY**

Proposed: Point names, controller references, meter IDs, physical tag captures and runtime signatures used only to corroborate identity. Coverage: All source systems involved in the selected engineering question, scoped to the relevant assets and systems. Timing: Current metadata snapshot plus recent point activity; summarize alias collisions, unmapped IDs and many-to-one mappings. Limits: Similar names do not prove identity; replacement equipment may retain old software tags, and one meter/point can legitimately serve multiple assets. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis + recommendation. Decision: Before joining datasets for diagnostics, maintenance, reporting or scenario analysis. Minimum evidence: At least two authoritative identifiers or one authoritative identifier plus verified physical mapping for every cross-system join used in a decision. Method: Normalize identifiers, compare location/parent system/manufacture context, detect one-to-many collisions, and label joins confirmed, candidate or rejected. Uncertainty: Behavioral similarity can falsely join separate assets; naming conventions change over time. Safe failure: Do not join candidate identities in decision outputs; return separate records and the exact owner/field check needed to resolve them.

WHY THIS IS USEFUL

The engineer can trust that multi-source trends refer to the same physical equipment and avoid false diagnoses caused by data-identity errors.

AUTHORITATIVE SUPPORTING DATA

Owners: Asset-information owner, BMS owner, metering owner, CMMS owner and discipline engineer. Precedence: Current authoritative asset ID/tagging convention and master asset record, then current controlled drawings and approved integration/point schedules; telemetry behavior is corroborative only. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Ambiguous aliases, duplicate IDs, retired tags still producing data, or inconsistent locations are conflicts that must remain visible until resolved by the owner. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not infer an asset identity only because two time series correlate, expose restricted controller/network details unnecessarily, or write inferred aliases back into source systems. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-008 Before intrusive work, which services are confirmed, probable or unknown within this ceiling, riser or plant area?**SENSORS AND TELEMETRY**

Proposed: No ambient telemetry is treated as proof of concealed services. Where authorised, verified detector/survey outputs or known local plant states may be referenced as supporting evidence only. Coverage: The exact work zone and adjacent service corridors likely to intersect the proposed work; avoid collecting unrelated room/person information. Timing: Use the latest authorised survey at the actual work location; telemetry is secondary and time-matched only if relevant. Limits: BMS/meter data cannot locate concealed cables, pipes or ducts. Historical drawings may omit alterations, and detection methods have material-specific limitations. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + calculation. Decision: At design development and immediately before drilling, cutting, opening-up or other intrusive work. Minimum evidence: Current service records for the work area plus the survey evidence required by the applicable safe system of work. Method: Overlay only georeferenced/current evidence; classify each service by source, date, location tolerance and confidence; identify no-go or further-survey zones without inventing routes. Uncertainty: Concealed routes can deviate between access points and detectors may miss shielded/non-metallic services. Safe failure: Return UNKNOWN where location cannot be proven and require the competent survey/opening-up procedure before intrusive work.

WHY THIS IS USEFUL

The engineer can define where further detection, opening-up or drawing correction is needed before detailing or permitting intrusive work.

AUTHORITATIVE SUPPORTING DATA

Owners: Project/design engineer, Estates record owner, permit-to-work owner and competent survey/maintenance teams. Precedence: Current controlled service drawings and latest approved intrusive/non-intrusive survey results, then competent-person site inspection and permit controls; the May 2021 floor plan is historical spatial context only. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Any drawing predating later works, unresolved survey limitation, access restriction or conflict between field evidence and drawing keeps the service location unknown. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not provide excavation/drilling clearance, certify an area free of services, infer concealed routes from adjacent rooms, or replace permit, survey and safe-working procedures. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

Drawings, asset registers and design intent

ME-009 Is there verified physical, service and maintenance-access capacity for the proposed M&E addition in this plantroom or riser?

SENSORS AND TELEMETRY

Proposed: Existing plant temperatures, electrical/thermal loads and runtime may inform service conditions; authorised dimensional scans or surveys can provide current geometry when available. Coverage: The candidate installation zone, access route, existing plant clearances and the services that constrain the addition. Timing: Current geometric survey and representative service-condition data; use peak/seasonal engineering summaries where capacity interaction matters. Limits: Telemetry cannot prove physical clearance, structural capacity, fire-stopping, lifting access or maintainability; scans can be occluded or outdated after works. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + estimate + recommendation. Decision: During option selection and before coordinated detailed design. Minimum evidence: Current dimensional evidence, proposed equipment envelope including access/maintenance clearances, verified service capacities and identified project constraints. Method: Compare envelopes and access paths; separately assess service headroom using current evidence; create a constraint list with required discipline checks. Uncertainty: A fit in geometric space does not establish installation feasibility, structural adequacy, heat rejection, code compliance or maintainability. Safe failure: Return an indicative constraint screen only; require coordinated design and competent surveys for unresolved interfaces.

AUTHORITATIVE SUPPORTING DATA

Owners: Project/design engineer, Estates space/asset owner, structural/fire/access specialists as applicable, and discipline information owners. Precedence: Current measured survey and approved current plantroom/riser records, then current service drawings and manufacturer maintenance/clearance requirements; historical plans are only reference. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Any post-survey alteration, uncertain service route, unavailable access zone or conflicting dimension is treated as unresolved. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not approve the installation, certify clearances/structure/fire performance, or infer inaccessible dimensions and service routes from the old floor plan. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can decide whether the proposal should proceed to detailed survey/design or be redirected because space, access or service constraints are unresolved.

ME-010 Which facts may use the May 2021 floor plan as historical context, and which present-day claims need current evidence?

SENSORS AND TELEMETRY

Proposed: No sensor can make a historical floor plan current. Verified asset/zone telemetry may reveal that a referenced space or service relationship needs contemporary confirmation. Coverage: Only the spatial areas relevant to the engineering question; telemetry is used to flag candidate changes, not to infer current use/access/services. Timing: Compare the plan issue date with current record effective dates and recent verified asset mappings; no present-tense state is derived solely from telemetry correlation. Limits: Room labels, partitions, access, service routes, hazards and closures may have changed; a live point in a named zone does not validate the floor-plan label. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + forecast + diagnosis. Decision: Whenever the 2021 plan is used to orient an engineering query, survey or scenario. Minimum evidence: The May 2021 plan plus a current authoritative source for every fact that affects present room use, access, service, hazard, closure or engineering condition. Method: Separate stable historical geometry from claims needing current confirmation; show source/effective dates side by side and suppress present-tense wording where evidence is missing. Uncertainty: Some geometry may remain unchanged but cannot be assumed unchanged because no later record is found. Safe failure: Use HISTORICAL CONTEXT ONLY or NOT VERIFIED and name the current source/survey required.

AUTHORITATIVE SUPPORTING DATA

Owners: Current space-information owner, Estates engineering record owners, project/change-control owners and relevant responsible persons. Precedence: Current authoritative space/engineering records and competent-person site evidence take precedence; the May 2021 plan is historical spatial reference only. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Any present-tense claim without a current owner-confirmed source is marked historical-only or not verified, even if the 2021 plan appears plausible. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not state that a room is currently used, accessible, open, safe or served by a system because the May 2021 plan says so. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can use the old plan for orientation without accidentally turning historical information into current engineering evidence.

ME-011 What is the evidenced maximum electrical demand, and what bounded headroom can be stated without treating it as design capacity?

SENSORS AND TELEMETRY

Proposed: Revenue/incomer meter active power, apparent power, current by phase and maximum-demand registers where available; transformer/incomer temperature may add context if mapped. Coverage: Building incomer and any upstream transformer/supply element for which the current rating and metering boundary are authoritative. Timing: Native interval data, preferably 15 min or finer, over at least 12 representative months where available; retain true interval maxima and seasonal peaks rather than monthly averages. Limits: Meter clipping, CT ratio errors, missing peaks, demand-window differences and embedded generation can distort apparent headroom. Measured demand is not cable, transformer or protection capacity. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + calculation + estimate + diagnosis + recommendation. Decision: At load-connection feasibility, annual capacity planning and before major electrification options. Minimum evidence: Verified metering boundary and scaling, current authoritative upstream ratings/constraints, and data covering representative seasonal/operational peaks. Method: Calculate interval maxima, P95/P99 and seasonal coincidence; subtract observed peak from only the specific authoritative operational limit being considered, while separately listing design checks still required. Uncertainty: Future coincidence, abnormal operating modes, harmonics, ambient conditions and protection/fault-level constraints can reduce usable capacity. Safe failure: If the observation period or authoritative limit is weak, report the measured peak only and state that usable electrical capacity remains NOT VERIFIED.

WHY THIS IS USEFUL

The engineer can establish whether a proposed load merits detailed design now, whether temporary monitoring is needed, and which upstream constraint must be checked first.

AUTHORITATIVE SUPPORTING DATA

Owners: Estates electrical duty holder/responsible person, metering/energy owner, DNO/supply information owner where relevant, and electrical design engineer. Precedence: Current supply agreement/connection data, transformer/incomer ratings and current controlled SLD/protection data, then validated meter configuration and competent-person tests; operational telemetry follows. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: If the metering boundary, CT ratio, transformer rating or supply arrangement is unresolved, maximum demand may be reported but headroom is not verified. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not call arithmetic difference "available capacity", approve a new load, state conductor/transformer adequacy, or omit voltage-drop, fault-level, discrimination, harmonic and protection checks required by design. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-012 Are any switchboards or distribution boards materially phase-imbalanced, and is neutral loading credible enough to investigate?

SENSORS AND TELEMETRY

Proposed: Per-phase RMS current, voltage and neutral current from verified multifunction meters or temporary power-quality instruments; phase-to-meter mapping must be known. Coverage: Selected incomers, switchboards and DBs with three-phase measurement; downstream single-phase circuits are included only when circuit metering is authoritative. Timing: 1 min or finer for diagnostic periods and 15 min summaries over representative weeks; retain coincident phase maxima and neutral-current peaks. Limits: Unequal CT accuracy, missing phases, harmonics and intermittent loads can misstate imbalance; neutral current can be harmonic-driven rather than simple load imbalance. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis. Decision: During recurring electrical assessment, unexplained overheating/nuisance issues or before redistributing single-phase loads. Minimum evidence: Simultaneous verified three-phase current data, neutral current where relevant, and current board/circuit mapping. Method: Calculate current imbalance as deviation from phase mean across comparable intervals, identify persistence and coincidence with neutral/harmonic behavior, and separate steady imbalance from short events. Uncertainty: Telemetry cannot identify the exact downstream circuit causing imbalance unless circuit-level evidence exists, and imbalance thresholds depend on equipment/system context. Safe failure: Flag boards for competent-person measurement and circuit investigation; do not prescribe phase moves from aggregate data.

WHY THIS IS USEFUL

The engineer can prioritize which boards need detailed load logging, thermal inspection or circuit redistribution design instead of surveying every board equally.

AUTHORITATIVE SUPPORTING DATA

Owners: Electrical duty holder/responsible person, metering owner and electrical engineer. Precedence: Current board schedules/SLD and protection/conductor records, then verified meter/CT configuration and competent-person power-quality measurements; BMS/energy data is supporting evidence. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Unknown CT orientation/ratio, changed circuit allocation or meter phase-label conflict makes the affected result indicative only. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not instruct live circuit redistribution, infer conductor temperature/condition from current alone, or certify neutral/cable adequacy. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-013 Which metered M&E systems are genuinely coincident with the building electrical peak rather than simply high annual consumers?**SENSORS AND TELEMETRY**

Proposed: Verified system submeters for HVAC plant, lighting or other authorised end-use groups plus building-incomer power. Coverage: Only approved aggregate end-use groups whose metering boundaries are mapped; no personal device or individual activity inference. Timing: 15 min or finer synchronized power for representative peak seasons; preserve peak-day traces and top-N coincident intervals. Limits: Submeter gaps, shared meters, nested meters and clock skew can double-count or misattribute contribution; unmetered loads remain residual, not assigned by guess. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + estimate + recommendation. Decision: For demand reduction, tariff/capacity planning and electrification option studies. Minimum evidence: Verified building meter, mapped system submeters, synchronized clocks and sufficient data to include multiple independent peak events. Method: Rank each metered group by contribution during the building top 10-20 demand intervals; show median/range across events and an explicit unmetered residual. Uncertainty: Correlation at peak does not prove controllability or causation, and annual kWh ranking can differ substantially from peak contribution. Safe failure: If meter reconciliation is poor, report only the reliably metered contributors and leave residual demand unallocated.

AUTHORITATIVE SUPPORTING DATA

Owners: Energy/metering owner, electrical engineer, asset-information owner and BMS owner for plant-state context. Precedence: Verified current meter hierarchy and scaling, current SLD/service mapping and authoritative end-use boundaries, then calibrated meter data; modelling assumptions are lower precedence. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Any ambiguous nesting, stale meter map or unexplained negative/residual balance is shown before attribution. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not infer individual laboratory, workstation or occupant behavior, promise peak reduction from shutting a system, or recommend operational curtailment without service/safety assessment. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can target demand-management or plant-sequencing studies at the systems that actually drive capacity peaks.

ME-014 Is there defensible evidence of voltage, frequency, unbalance or harmonic power-quality problems at critical distribution points?**SENSORS AND TELEMETRY**

Proposed: Class-appropriate power-quality measurements: RMS voltage, frequency, voltage unbalance, THD and event captures for sags/swells/interruptions; BMS meters are used only if their capabilities are documented. Coverage: Incomer and nominated critical boards/equipment connection points; temporary instruments require authorised installation. Timing: Cycle/event-resolution capture for disturbances and 10 min or instrument-standard aggregation for trends over a representative monitoring period. Limits: Many energy meters do not meet diagnostic PQ performance; THD without harmonic spectrum or event waveform may be insufficient for root-cause work. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis + recommendation. Decision: When equipment trips/resets recur, complaints arise, or sensitive loads are planned. Minimum evidence: Verified measurement point, instrument specification/calibration, synchronized event record and plant/switching context for representative incidents. Method: Classify event type/magnitude/duration, compare simultaneous upstream/downstream captures where available, and test association with known switching without asserting causation from timing alone. Uncertainty: A single measurement point rarely distinguishes upstream supply disturbance from downstream load interaction. Safe failure: State the observed PQ phenomenon and confidence, then specify additional simultaneous logging or competent-person investigation required.

AUTHORITATIVE SUPPORTING DATA

Owners: Electrical duty holder/responsible person, electrical engineer, metering owner and supply/DNO contact where relevant. Precedence: Competent-person power-quality test results and current supply/protection information, then verified instrument configuration/calibration and fixed-meter data; generic BMS values are lower precedence. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Results outside instrument capability, expired calibration, unknown connection point or clock mismatch are not used for a PQ conclusion. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not certify supply quality, blame the DNO or a specific load from correlation alone, alter protection, or advise live testing outside competent procedures. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can decide whether a power-quality survey, load investigation or protection/design assessment is justified and where instruments should be placed.

Electrical distribution, loads and resilience

ME-015 Which loads are verified on UPS, generator or other resilient supplies, and what evidence exists for ride-through or autonomy?

SENSORS AND TELEMETRY

Proposed: Read-only UPS input/output power, battery state-of-charge/health indicators, generator status, ATS position feedback and supported-board meters where present. Coverage: Each resilience source, transfer device and documented downstream critical board/load group; uninstrumented branches stay unknown. Timing: Continuous 1-60 s states for transfer events plus 5-15 min load/autonomy trends; use records from authorised tests or real outages, not deliberate switching by the service. Limits: ATS indication may not prove every downstream load remained functional; battery vendor estimates and SOC are not guaranteed autonomy under all conditions. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + forecast. Decision: Before critical-work planning, resilience exercises, load additions or business-continuity engineering. Minimum evidence: Current topology/critical-load schedule plus latest approved functional/load test and verified operating telemetry for the relevant source. Method: Reconcile mapped supported loads, quantify observed source loading during tests/events, compare available rated/tested capacity and identify branches with no ride-through evidence. Uncertainty: Rated UPS/generator capacity does not guarantee duration, fuel availability, battery condition or downstream equipment restart behavior. Safe failure: Report supported-by-record, supported-by-test, conflicted or unverified; require authorised test evidence for any critical claim.

AUTHORITATIVE SUPPORTING DATA

Owners: Electrical duty holder/responsible person, resilience/critical-services owner, maintenance/test owner and electrical engineer. Precedence: Current resilient-supply SLD and critical-load schedule, approved cause/effect/transfer design and latest competent-person load-bank/functional test records, then verified UPS/generator telemetry. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Any undocumented load transfer, expired battery/test evidence, altered critical-load schedule or inconsistent ATS/board mapping is flagged. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not guarantee continuity, battery runtime, generator start or transfer success, initiate resilience tests, or declare a load "critical" without the authoritative owner. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can expose resilience mapping gaps, judge whether a proposed critical load needs a new capacity/test study and prepare an evidence-based test scope.

ME-016 What is the most defensible timeline for the last outage, voltage disturbance or unexplained plant reset?

SENSORS AND TELEMETRY

Proposed: Incomer/board voltage and power, breaker/ATS/UPS events, BMS controller restarts, plant enable/feedback and network-free device uptime indicators where legitimately available. Coverage: The affected electrical path and directly dependent plant/control assets only; no inspection of user network traffic or research data. Timing: Use native event timestamps at highest available resolution with 15 min pre/post context; record clock offsets before ordering events. Limits: Different systems buffer events, quantize timestamps or reboot with clock drift; absence of telemetry during power loss is itself ambiguous. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + calculation + estimate + diagnosis. Decision: For each material outage/reset incident, as soon as evidence is preserved. Minimum evidence: At least one authoritative event record, verified topology, clock-offset estimates and two independent evidence streams for key causal transitions where possible. Method: Normalize clocks, create earliest/latest time bounds for each event, distinguish observed facts from inferred sequence, and test alternative explanations such as upstream dip versus local trip. Uncertainty: Temporal order supports hypotheses but does not prove root cause, especially with coarse or drifting clocks. Safe failure: Present a confidence-coded timeline and unresolved alternatives; require protection/PQ or physical investigation where causation remains uncertain.

AUTHORITATIVE SUPPORTING DATA

Owners: Electrical duty holder, BMS/controls owner, incident/maintenance record owner and affected service owner. Precedence: Competent-person incident/switching records and protection/UPS logs, then current topology and verified event telemetry; help-desk reports are contextual and may have imprecise times. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Clock uncertainty, missing logs, post-event configuration changes or unverified device mappings are carried into the timeline. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not claim a root cause from timing alone, disclose unrelated user/network activity, reset equipment, reclose breakers or alter protection settings. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can preserve a coherent incident record and choose the next diagnostic measurement instead of troubleshooting from anecdotal timing.

ME-017 Which feeders or boards show sustained loading that warrants detailed cable, protection and capacity checks?**SENSORS AND TELEMETRY**

Proposed: Per-phase current, demand and power-factor measurements at mapped feeders/boards; temperature sensing only where installed for that purpose and properly located. Coverage: Selected feeders and boards with authoritative upstream/downstream identity; unknown downstream diversity is not invented. Timing: 1-15 min load data covering representative seasonal peaks and known high-load operating modes; report sustained percentile and maxima separately. Limits: Breaker rating is not conductor capacity; thermal environment, grouping, harmonic neutral current, voltage drop and installation method are not inferable from meter current. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + forecast + recommendation. Decision: Before adding load, after recurring overcurrent/thermal concerns, or during capacity planning. Minimum evidence: Verified current measurements plus current cable/protection design data for the exact feeder and representative peak data. Method: Calculate sustained and peak utilization against the documented design current/rating while keeping separate flags for cable derating, voltage drop, harmonics and discrimination checks. Uncertainty: Observed historical maxima do not bound future demand and may miss abnormal combinations. Safe failure: Rank feeders for detailed competent-person design/calculation; never infer safe spare capacity where design inputs are incomplete.

WHY THIS IS USEFUL

The engineer can focus detailed cable/protection calculations on feeders where measured utilization makes the question decision-relevant.

AUTHORITATIVE SUPPORTING DATA

Owners: Electrical duty holder/responsible person, electrical design record owner and metering owner. Precedence: Current cable schedules/design calculations, protective-device settings, board schedules and installation records, then competent-person inspection/test data; verified load telemetry is an operational input. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: If conductor size/method, protection setting, meter scaling or circuit identity is uncertain, utilization can be reported but capacity margin cannot. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not equate breaker amperes with usable feeder capacity, approve load additions, state cable thermal safety or protection coordination, or direct protective-device changes. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-018 Can this location plausibly support a proposed high-load item, and which electrical design checks remain mandatory?**SENSORS AND TELEMETRY**

Proposed: Local DB/submeter demand, per-phase current, voltage and upstream incomer demand where mapped; optional PQ logging if the proposed load is nonlinear/sensitive. Coverage: Candidate connection board, upstream feeder path and building supply boundary; nearby meters are not treated as the candidate circuit unless mapping is proven. Timing: Representative 15 min demand over peak seasons plus short-interval logging where motor/inrush or rapidly varying loads matter. Limits: Historical spare demand does not establish fault level, voltage drop, starting performance, harmonic compatibility, protection selectivity or physical spare ways. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis + recommendation. Decision: At option selection before detailed electrical design or procurement. Minimum evidence: Verified candidate connection path, equipment steady/inrush/nonlinear load data, representative measured demand and current upstream design constraints. Method: Compare observed coincident demand with documented limits, identify phase/peak issues and enumerate required design calculations/tests: cable, protection, voltage drop, fault level, discrimination, harmonics and resilience as applicable. Uncertainty: Future coincidence and equipment behavior can differ from nameplate assumptions; meter data cannot close design-code checks. Safe failure: Return PLAUSIBLE / CONSTRAINED / NOT VERIFIED with the exact detailed-design evidence required, not an approval.

WHY THIS IS USEFUL

The engineer can eliminate unsuitable connection options early and commission the correct detailed calculations and surveys for the viable location.

AUTHORITATIVE SUPPORTING DATA

Owners: Project electrical engineer, electrical duty holder/responsible person, Estates electrical records owner and metering owner. Precedence: Approved equipment load characteristics and current design brief, current SLD/board/cable/protection data and supply constraints, then competent-person site/test evidence; telemetry provides observed baseline only. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: If the proposed load profile, connection point or any upstream design record is provisional, feasibility remains indicative. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not approve the connection, allocate a circuit, guarantee power quality or resilience, or substitute a telemetry screen for electrical design and competent-person verification. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

Electrical distribution, loads and resilience

ME-019 How large is the persistent electrical baseload, and which verified M&E end-use groups account for it?

SENSORS AND TELEMETRY

Proposed: Building-incomer power plus mapped aggregate submeters for HVAC, lighting and other authorised M&E groups. Coverage: Whole-building supply and only end-use groups with validated meter boundaries; no socket/device/person attribution unless already an approved aggregate engineering meter. Timing: 5-15 min data across at least several normal weeks in different seasons; define baseload from lower-load quantiles and stable out-of-hours windows rather than a single nightly minimum. Limits: Scheduled 24/7 plant, research equipment, meter nesting and embedded generation can distort residual baseload; absence of submetering prevents attribution. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + estimate + recommendation. Decision: Monthly/seasonal energy assessment and before controls or shutdown optimization studies. Minimum evidence: Verified incomer data, synchronized aggregate submeters covering a meaningful share of demand, and known current operating schedules for major 24/7 services. Method: Estimate robust lower-load envelope by season/day type, reconcile metered groups to the incomer and show unallocated residual; test persistence before naming a candidate source. Uncertainty: A persistent metered load does not imply waste; it may support safety, research or environmental requirements. Safe failure: If reconciliation is weak, report baseload magnitude and metered contributors only, leaving the residual unexplained.

AUTHORITATIVE SUPPORTING DATA

Owners: Energy/metering owner, electrical engineer, BMS operator and relevant service owners. Precedence: Verified current meter hierarchy and schedules/plant operating requirements, then validated meter data and current operational records; inferred residual categories have lower confidence. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Changes in operating regime, meter topology or missing intervals are segmented rather than blended into one baseline. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not identify occupants, personal devices or research equipment from load signatures, label a load wasteful without service context, or command shutdowns. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can decide which aggregate end-use groups deserve plant/schedule investigation and quantify the unmetered evidence gap.

ME-020 What are the credible service consequences if one electrical supply, board or resilience component becomes unavailable?

SENSORS AND TELEMETRY

Proposed: Verified breaker/ATS/UPS states, board loads and dependent plant operating states used to corroborate documented dependencies. Coverage: The nominated failure element and mapped downstream electrical/mechanical services; unverified branches are shown as unknown rather than assumed. Timing: Current load state plus evidence from authorised past outages/tests; use scenario snapshots at normal and peak operating conditions. Limits: A mapped dependency does not prove service failure mode: local UPS, restart logic, stored thermal capacity or alternative feeds may alter consequences. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + forecast + recommendation. Decision: For resilience planning, planned shutdown preparation and design option comparison. Minimum evidence: Current verified topology, service dependency map, current load state and test/incident evidence for claimed resilience behavior. Method: Propagate component unavailability through documented dependencies, separate immediate electrical loss from delayed service effects, and compare normal/peak scenarios with confidence flags. Uncertainty: N-1 modeling cannot prove actual transfer, restart or autonomy under the future event. Safe failure: Return a scenario with unknown dependencies highlighted and specify which authorised resilience test or field verification is needed.

AUTHORITATIVE SUPPORTING DATA

Owners: Electrical duty holder/responsible person, critical-services/business-continuity owner, discipline engineers and approved resilience-test record owner. Precedence: Current SLD/resilience topology, critical-service dependency records and latest authorised functional tests, then verified operational telemetry; historical incident reports are contextual. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Any untested transfer path, undocumented local resilience or changed critical-load mapping is carried as an uncertainty, not filled by assumption. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not declare a service resilient, guarantee N-1 performance, initiate transfer tests, advise switching sequences or expose restricted critical-infrastructure detail beyond authorised roles. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can plan shutdown sequencing, identify single points needing test/mitigation and challenge resilience assumptions before a live event.

ME-021 Is this AHU delivering the air conditions and operating states expected for the zones it is verified to serve?**SENSORS AND TELEMETRY**

Proposed: AHU enable/fan status, supply/return/outdoor-air temperature and humidity, static pressure, airflow where measured, damper/valve feedback and mapped zone temperature/CO2. Coverage: The AHU, its verified sensor locations and only zones with current service mapping. Timing: 10-60 s raw states with 5 min steady-state summaries across representative occupied/unoccupied and weather conditions. Limits: Sensor placement and mixed-air stratification can bias values; fan status does not prove airflow, damper position does not prove outdoor-air fraction, and CO2 does not directly measure airflow. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis. Decision: During performance assessment, complaint diagnosis, recommissioning or change-of-use work. Minimum evidence: Current serving-zone map, approved control/design intent, verified sensor chain and representative periods at meaningful load/weather. Method: Segment operating modes, compare supply conditions and pressure/airflow with intended reset logic, and relate downstream zone indicators without treating correlation as proof of airflow adequacy. Uncertainty: Duct leakage, terminal control, sensor bias and zone loads can produce similar symptoms. Safe failure: Report deviations and ranked hypotheses; require airflow/balancing or physical tests where the decision depends on actual ventilation delivery.

AUTHORITATIVE SUPPORTING DATA

Owners: Mechanical engineer, BMS owner/operator, current zoning/space-information owner and commissioning/balancing record owner. Precedence: Current AHU design/control intent and verified zone-service mapping, latest commissioning/balancing evidence and current setpoints/schedules, then verified BMS telemetry. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Changed zone use, altered ductwork, unresolved point mapping or stale balancing data limits comparisons to observed plant behavior only. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not certify ventilation adequacy, infer room occupancy from CO2, alter AHU setpoints/dampers, or claim a zone is served without current mapping. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can distinguish an AHU-level control/supply issue from a downstream distribution problem and choose the next measurement accordingly.

ME-022 When a heating or cooling coil underperforms, does evidence point to coil, water-side, air-side or control limitations?**SENSORS AND TELEMETRY**

Proposed: Coil entering/leaving air and water temperatures, valve command/feedback, water flow or differential pressure where measured, fan/airflow indicators and plant supply temperature. Coverage: The coil, its control valve, relevant hydronic branch and AHU air path. Timing: 10-60 s data through several sustained high-demand periods; calculate stable 5-15 min windows and retain valve/flow transitions. Limits: Valve position is not water flow; sensor placement, bypass leakage, fouling, low primary temperature or low airflow can mimic capacity saturation. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis + recommendation. Decision: During repeated high-demand failure or before coil/valve/plant replacement. Minimum evidence: Verified temperatures, valve state, at least one credible water-flow/DP indicator, air-side context and current design/control intent during sustained demand. Method: Compare approach temperatures, water delta-T, air delta-T, control saturation and upstream supply; test competing hypotheses rather than choosing the largest deviation. Uncertainty: Without measured airflow/water flow, delivered duty has wide uncertainty and multiple constraints may coexist. Safe failure: Return the best-supported constraint class plus the temporary instrument or physical check needed to discriminate alternatives.

AUTHORITATIVE SUPPORTING DATA

Owners: Mechanical engineer, commissioning/balancing owner, BMS owner and plant maintenance owner. Precedence: Current coil selection/design duty and approved control sequence, current hydronic/air balancing data and competent-person measurements, then verified telemetry. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: If coil/valve has been replaced, design duty is uncertain, or flow sensors are not valid, the root-cause classification remains indicative. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not declare coil condition, fouling or valve failure solely from BMS trends, size replacement coils, or command valves/pumps. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can avoid replacing the wrong component and target balancing, valve, plant-temperature or airflow investigation first.

ME-023 Are VAV boxes or dampers distributing air in line with current design intent, and which zones need airflow verification?**SENSORS AND TELEMETRY**

Proposed: Terminal airflow where a commissioned flow station exists, damper command/feedback, zone temperature/CO₂, AHU static pressure and fan speed. Coverage: All terminals on the selected branch plus the controlling static-pressure sensor and representative zones. Timing: 10-60 s states with 5 min summaries over occupied periods and high/low-load conditions; preserve damper saturation episodes. Limits: Damper position is not airflow; flow pickups drift with fouling/installation, and CO₂ cannot distinguish low airflow from high generation or mixing. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis + recommendation. Decision: During zone complaint clusters, recommissioning or before changing static-pressure/reset strategy. Minimum evidence: Verified terminal identities, current design/balancing targets and reliable flow feedback for enough terminals to represent the branch. Method: Compare flow error, damper saturation, branch static pressure and simultaneous terminal demand; identify starved/over-supplied patterns and critical-path candidates. Uncertainty: A terminal can report correct airflow while diffuser distribution remains poor; unmeasured leakage or duct restrictions may be upstream. Safe failure: Flag terminals/branches for competent airflow measurement; do not treat proxy-only zones as balanced.

AUTHORITATIVE SUPPORTING DATA

Owners: Mechanical engineer, BMS owner, current terminal/zone asset mapping owner and balancing/commissioning record owner. Precedence: Current ventilation design airflow/terminal schedule and latest balancing results, then approved control sequence and verified BMS point mapping; telemetry follows. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Changed terminal assignment, stale K-factors or unresolved sensor calibration makes calculated airflow unsuitable for acceptance. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not certify balancing, infer occupancy from terminal airflow/CO₂, or change minimum airflow and pressure setpoints automatically. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can choose the smallest set of terminal measurements and branch checks needed to verify distribution rather than rebalance blindly.

ME-024 Is the AHU heat-recovery section delivering useful recovery, and do bypass or frost-control states explain poor performance?**SENSORS AND TELEMETRY**

Proposed: Outdoor, return, supply and exhaust air temperatures, heat-recovery wheel/plate run or bypass position, fan states and humidity where relevant. Coverage: The heat-recovery section and sensor locations immediately upstream/downstream of the exchanger where physically verified. Timing: 1 min or finer during periods with sufficient indoor-outdoor temperature difference; calculate steady 5-15 min effectiveness windows and exclude start-up/transient periods. Limits: Sensor offset strongly biases effectiveness; recirculation, leakage, frost protection and unequal airflows can invalidate simple temperature-effectiveness calculations. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + estimate + diagnosis. Decision: Seasonally and after maintenance/control changes. Minimum evidence: Verified air-temperature sensors, current exchanger/bypass control intent, confirmed fan operation and multiple stable periods with adequate temperature driving force. Method: Calculate temperature effectiveness with uncertainty bounds, stratify by bypass/frost mode and compare like-for-like outdoor conditions; identify persistent low-effectiveness states. Uncertainty: Temperature effectiveness is not total enthalpy recovery and cannot diagnose leakage/fouling without further tests. Safe failure: Return an indicative effectiveness range and specify sensor validation, airflow test or physical inspection before maintenance conclusions.

AUTHORITATIVE SUPPORTING DATA

Owners: Mechanical engineer, BMS owner, AHU commissioning owner and maintenance owner. Precedence: Current heat-recovery design/control intent, exchanger type/selection and latest commissioning/maintenance evidence, then verified sensor data. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: If sensor locations/calibration, bypass semantics or airflow balance are unknown, performance is reported as an indicator only. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not certify heat-recovery efficiency, infer contamination/leakage, bypass frost controls, or override recovery modes. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can determine whether poor recovery is likely measurement/control-state related or merits exchanger inspection and recommissioning.

ME-025 Is fan electrical demand proportionate to delivered airflow and pressure duty, or does the air system need investigation?**SENSORS AND TELEMETRY**

Proposed: Fan electrical power, speed/frequency, duct static pressure and measured airflow where a valid station exists; filter differential pressure and damper positions may add context. Coverage: The selected fan/AHU and the verified airflow/pressure measurement stations defining its duty. Timing: 10-60 s samples with 5-15 min steady-state windows across a range of fan speeds and system demand. Limits: Without reliable airflow the result is not specific fan power; VSD losses, motor efficiency, sensor placement and parallel fan operation affect interpretation. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis + recommendation. Decision: During energy assessment, persistent high pressure/damper saturation or before fan/VSD retrofit. Minimum evidence: Verified fan power plus credible airflow and/or pressure duty at several stable operating points and current design/control information. Method: Plot power against speed/airflow/pressure, compare repeatable operating points, and identify rising resistance or control patterns; use uncertainty bands rather than a single efficiency figure. Uncertainty: Higher power can reflect legitimately higher duty; telemetry cannot separate fan condition from system resistance without physical checks. Safe failure: If airflow is unavailable, report power/speed/pressure behavior only and recommend temporary airflow measurement before efficiency claims.

WHY THIS IS USEFUL

The engineer can decide whether to inspect filters/dampers/duct resistance, verify airflow sensing or pursue fan/VSD optimization.

AUTHORITATIVE SUPPORTING DATA

Owners: Mechanical engineer, energy/metering owner, BMS owner and commissioning/balancing owner. Precedence: Current fan selection/design duty and approved control strategy, latest airflow/pressure commissioning data and verified power-meter configuration, then telemetry. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Changed pulleys/fans/filters/ductwork, uncertain meter boundary or stale airflow calibration makes benchmark comparison indicative. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not claim fan efficiency, duct cleanliness or filter condition from electrical power alone, or change VSD/pressure settings. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-026 Is this heating or chilled-water circuit delivering expected flow and temperature difference, and where is hydraulic constraint most likely?**SENSORS AND TELEMETRY**

Proposed: Supply/return water temperatures, branch flow where metered, differential pressure, pump speed and valve command/feedback at selected critical branches. Coverage: Plant headers, the selected distribution branch and known critical/remote terminals. Timing: 10-60 s data with 5-15 min stable load windows across different pump speeds and load levels; preserve simultaneous valve saturation states. Limits: Temperature delta without flow does not give duty; bypasses, mixing, sensor placement and low load can create low delta-T without a fault. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis + recommendation. Decision: During low delta-T, poor terminal duty, pump-energy concerns or recommissioning. Minimum evidence: Current design flow/DP targets, verified temperature sensors, credible flow or DP evidence and simultaneous pump/valve states under meaningful load. Method: Calculate flow/delta-T/duty where valid, identify pump/valve saturation and compare remote branch behavior; test low-flow, excess-flow and mixing hypotheses. Uncertainty: Aggregate header data can mask branch problems and heat-transfer performance can limit delta-T independently of hydraulics. Safe failure: Return a branch-level hypothesis map and the balancing/temporary flow measurement needed to confirm it.

WHY THIS IS USEFUL

The engineer can target the right branch for flow verification, balancing or control investigation instead of treating low delta-T as a plant-only problem.

AUTHORITATIVE SUPPORTING DATA

Owners: Mechanical engineer, hydronic commissioning/balancing owner, BMS owner and plant maintenance owner. Precedence: Current schematic/design flow and pump/valve strategy, latest balancing/commissioning results and competent-person flow/DP measurements, then verified telemetry. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Unverified branch topology, changed balancing valves, sensor drift or unknown bypass configuration keeps hydraulic conclusions indicative. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not prescribe balancing valve positions, certify design flow, infer pipe condition, or change pump setpoints/valves. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-027 Is the pump differential-pressure strategy satisfying the critical branch without excessive pumping or throttling?**SENSORS AND TELEMETRY**

Proposed: Pump speed and electrical power, differential pressure at plant and remote sensors, representative valve positions and branch flow where available. Coverage: The selected variable-flow circuit from pump to the hydraulically critical branches with current sensor mappings. Timing: 10-60 s data over low, medium and high-load periods; summarize 5 min valve saturation and DP margin distributions. Limits: A remote DP sensor may not be at the true critical branch; valve position can be mis-scaled and high throttling may reflect terminal/control faults. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + calculation + diagnosis + recommendation. Decision: For control optimization, pump-energy investigation and after terminal/branch changes. Minimum evidence: Verified pump/DP/valve mappings, current control sequence and representative load periods that include near-peak demand. Method: Assess whether one or more legitimate terminal valves saturate while DP margin exists elsewhere; relate pump power/speed to DP target and distinguish systematic from local constraints. Uncertainty: Valve saturation is not always hydraulic starvation; coils/actuators/sensors may be faulty and branch flows may be unknown. Safe failure: Identify candidate control or balancing tests only; require competent recommissioning before setpoint changes.

AUTHORITATIVE SUPPORTING DATA

Owners: Mechanical engineer, BMS controls owner, commissioning/balancing owner and pump maintenance owner. Precedence: Current approved DP/reset control strategy, hydronic design/balancing data and current critical-branch definition, then verified pump/DP/valve telemetry. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: If critical branch, sensor location or control logic has changed, observed DP cannot be judged against historical intent without revalidation. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not alter DP setpoints, VSD limits or valve positions, and do not promise energy savings before confirming service performance. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can identify whether a DP-reset study or branch balancing is likely to reduce pump energy without compromising critical flow.

ME-028 Are heating and cooling being delivered simultaneously beyond what the approved control intent requires?**SENSORS AND TELEMETRY**

Proposed: Heating/cooling valve commands and feedback, coil temperatures, supply-air temperature, zone temperature, reheat state and plant enable signals. Coverage: The selected AHU/terminal/zone path with verified serving relationships; shared systems are kept separate from zone-level conclusions. Timing: 10-60 s states summarized into 5 min overlap episodes across representative seasons and operating modes. Limits: Valve command overlap does not prove energy transfer; leaking valves, minimum reheat, humidity control or heat-recovery logic can create legitimate overlap. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + recommendation. Decision: Monthly/seasonal controls assessment and when simultaneous plant demand is suspected. Minimum evidence: Current approved sequence, verified valve/coil/zone points and enough seasonal data to distinguish intentional humidity/frost/reheat modes. Method: Detect sustained simultaneous demands/thermal evidence, classify by control mode and quantify duration/energy only where meter/duty evidence is credible. Uncertainty: Without flow and temperature data, energy magnitude is uncertain and overlap may be commanded but not delivered. Safe failure: Flag episodes that conflict with current intent and require controls/valve functional testing before corrective action.

AUTHORITATIVE SUPPORTING DATA

Owners: Mechanical engineer, BMS controls owner and current control-sequence/design owner. Precedence: Current approved sequence/setpoints and zone/terminal mapping, then commissioning evidence and verified telemetry; generic energy rules do not override project-specific intent. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Unknown valve action, changed sequence or missing coil temperature evidence makes overlap a control-state indicator only. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not label all overlap wasteful, disable reheat/frost/humidity functions, change deadbands or infer valve leakage without testing. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can distinguish intended sequence behavior from candidates for valve, sensor or controls recommissioning.

ME-029 What evidence exists for ventilation delivery in this zone, and which claims still require direct airflow or balancing measurements?**SENSORS AND TELEMETRY**

Proposed: Terminal/branch airflow only where commissioned flow stations are valid; AHU airflow, damper state, fan pressure/speed and zone CO2 may be used as supporting indicators. Coverage: The nominated zone, its verified terminal(s) and serving AHU/branch; no person counting unless an approved anonymous aggregate sensor is separately verified. Timing: 1-5 min environmental data and 10-60 s plant states over representative occupied periods; direct airflow measurements use their test timestamps and conditions. Limits: CO2 is influenced by generation, mixing, outdoor CO2 and occupancy and is not a direct L/s measurement; fan/damper state cannot prove zone airflow. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis + recommendation. Decision: For complaint investigation, change of use, ventilation assurance and recommissioning planning. Minimum evidence: A current airflow requirement, verified terminal/service mapping and either current direct airflow evidence or explicit acknowledgement that only proxies exist. Method: Separate direct airflow evidence from indirect indicators; compare direct measurements with design criteria when valid, and use CO2 only to flag periods needing investigation. Uncertainty: Spatial mixing and variable occupancy make a single CO2 sensor non-representative; historical balancing may not reflect current system condition. Safe failure: Return DIRECTLY SUPPORTED, INDICATIVE or NOT VERIFIED and specify the airflow test locations needed for a direct conclusion.

WHY THIS IS USEFUL

The engineer can distinguish ventilation indicators from actual delivered-air evidence and target the exact airflow or balancing measurements still required.

AUTHORITATIVE SUPPORTING DATA

Owners: Mechanical engineer, current space/use owner, ventilation commissioning/balancing owner and BMS owner. Precedence: Current ventilation design criteria and zone-service mapping, latest competent-person airflow/balancing results and current room/use brief, then verified telemetry as trend/context. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: If current use, design airflow target or terminal mapping is unresolved, adequacy cannot be judged from environmental trends. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not certify ventilation adequacy, infer exact occupancy from CO2, make health claims, or substitute telemetry for required airflow testing. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-030 For a persistent hot or cold zone, which M&E causes are best supported and which still need field testing?**SENSORS AND TELEMETRY**

Proposed: Zone/supply temperatures, terminal airflow if valid, damper/valve feedback, heating/cooling availability, AHU states and nearby outdoor-condition data. Coverage: The affected zone plus its verified terminal, branch and upstream plant; compare nearby peer zones only when they share system context. Timing: 1-5 min zone data and 10-60 s plant states across multiple complaint periods and comparable non-complaint periods. Limits: A single zone sensor may be poorly located; solar/internal gains, open doors or local equipment may be unmeasured and must not be attributed to occupants. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + estimate + diagnosis. Decision: After repeated complaints or sustained criterion excursions, before intrusive investigation. Minimum evidence: Verified zone/service mapping, representative complaint/non-complaint windows, current control intent and sufficient plant/terminal evidence to test at least two plausible causes. Method: Compare sensor bias, airflow limitation, valve/coil response, schedule, supply condition and plant availability; use peer-zone comparison cautiously and rank hypotheses by evidence. Uncertainty: Unmeasured local gains/envelope effects can produce identical symptoms and telemetry rarely proves physical failure. Safe failure: Return ranked hypotheses with discriminating tests; do not state a root cause until direct evidence supports it.

WHY THIS IS USEFUL

The engineer can choose the next field check with the highest diagnostic value and avoid replacing components based on a temperature trend alone.

AUTHORITATIVE SUPPORTING DATA

Owners: Mechanical engineer, BMS owner, maintenance owner and current room/service mapping owner. Precedence: Current design/control intent and zone-service mapping, recent maintenance/commissioning evidence and competent-person observations, then verified telemetry. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: If complaint timing, current room use or serving system is uncertain, root-cause ranking is weakened and must say so. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not infer occupant behavior or blame users, alter comfort setpoints automatically, or diagnose valve/damper/coil failure without appropriate testing. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-031 What is the current building water-use baseline, and how much reconciles to verified submetered branches?**SENSORS AND TELEMETRY**

Proposed: Building main water meter and mapped branch/submeters with pulse/volume/rate data; plant make-up meters where independently mapped. Coverage: Whole-building water boundary and authorised engineering branch groups only; no individual fixture/user attribution. Timing: Native pulse/interval data aggregated to 15 min, daily and weekly totals across representative months; retain meter resets and gaps. Limits: Pulse multipliers, rollover, reverse flow, nested meters and clock gaps can break reconciliation; meter totals do not reveal purpose of use. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis. Decision: Monthly/seasonal water assessment, leak investigation preparation and retrofit baselining. Minimum evidence: Verified main meter, mapped submeter hierarchy, synchronized interval data and enough coverage to include normal operating variation. Method: Reconcile nested meters, quantify mapped consumption and residual with uncertainty, segment by day type/season, and identify abrupt baseline changes without assigning cause prematurely. Uncertainty: Unmetered branches and meter accuracy limit the balance; process changes can alter baseline without leakage. Safe failure: Report reconciled share and unresolved residual; require meter verification or additional submetering before branch attribution.

AUTHORITATIVE SUPPORTING DATA

Owners: Estates water/utility owner, metering owner, mechanical/public-health engineer and asset-information owner. Precedence: Current utility/main-meter boundary and verified meter hierarchy/scaling, then approved branch schematics and competent-person meter checks; analytics-derived residuals are lower confidence. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Any uncertain multiplier, shared branch, meter replacement or unexplained negative balance is surfaced before allocation. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not infer individual water use, declare a leak solely from unreconciled volume, or treat utility billing data as proof of internal branch allocation. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can establish whether the water metering architecture is good enough for leak, demand or efficiency decisions and identify the missing branch evidence.

ME-032 Is persistent out-of-hours water flow strong enough to justify a targeted leak or uncontrolled-flow investigation?**SENSORS AND TELEMETRY**

Proposed: Main and mapped branch water flow/volume meters, plus known plant make-up/irrigation/process meter states where authorised. Coverage: Building and relevant branch meters only; no fixture-level personal behavior inference. Timing: 1-15 min flow across multiple nights/weekends and known low-demand periods; report minimum stable flow, duration and recurrence. Limits: Legitimate 24/7 processes, tank filling, flushing and meter pulse quantization can mimic leakage; a zero-reading threshold may miss small leaks. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis + recommendation. Decision: Weekly/monthly screening and after unexplained consumption increases. Minimum evidence: Verified meter data across several comparable low-demand windows plus current authorised information on legitimate continuous/intermittent water uses. Method: Identify recurrent stable flow, subtract only metered/authoritatively scheduled uses, compare branch patterns and rank likely investigation zones by evidence. Uncertainty: Unmetered legitimate uses and intermittent leaks create false positives/negatives. Safe failure: Return INVESTIGATE / INCONCLUSIVE with the branch isolation/leak-detection test needed; never declare leak location from aggregate flow.

AUTHORITATIVE SUPPORTING DATA

Owners: Water/public-health engineer, Estates utility owner, maintenance owner and relevant process/service owners. Precedence: Current known operating schedules/process water requirements and verified meter hierarchy, then competent-person inspection/leak-test evidence; interval telemetry is a screening source. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: If out-of-hours operating requirements or meter sensitivity are unknown, leakage cannot be asserted. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not infer who used water, direct valve isolation, declare premises leak-free, or replace physical leak detection and competent inspection. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can decide whether an overnight flow warrants targeted branch testing and avoid unnecessary intrusive leak searches.

ME-033 Is the domestic hot-water plant recovering and circulating as intended, without turning telemetry into a hygiene-compliance claim?**SENSORS AND TELEMETRY**

Proposed: DHW calorifier/plant flow and return temperatures, recirculation pump status, branch return temperatures where instrumented and heat input where measured. Coverage: DHW generation and recirculation loops with verified sensor locations; no fixture-user monitoring. Timing: 1 min to 5 min data through draw-off recovery and low-demand periods; summarize recovery time, return-temperature patterns and pump availability. Limits: Fixed sensors may not represent distal outlets, calibration matters around control thresholds, and return temperature alone does not establish every outlet condition. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + calculation + diagnosis. Decision: Operational screening, complaint/fault diagnosis and preparation for competent water-hygiene work. Minimum evidence: Current system schematic/control intent, verified plant/return sensors and representative recovery/low-demand cycles; hygiene decisions additionally require the authoritative control regime and competent-person checks. Method: Measure recovery duration, plant/return temperature trajectories and pump-state consistency; flag slow recovery, intermittent circulation or unstable control for investigation. Uncertainty: Telemetry does not cover all outlets, stagnation, dead legs or microbiological risk and cannot substitute for the control scheme. Safe failure: Report plant/circulation performance only and refer hygiene significance to the responsible competent person.

WHY THIS IS USEFUL

The engineer can identify DHW plant or circulation behavior that deserves physical investigation while keeping hygiene decisions inside the competent control regime.

AUTHORITATIVE SUPPORTING DATA

Owners: Responsible person for water hygiene, public-health/mechanical engineer, water-hygiene control-regime owner and maintenance owner. Precedence: Current water-hygiene control scheme/risk assessment and competent-person temperature/inspection records, current DHW design/commissioning information, then verified telemetry as supplementary trend evidence. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Any expired risk/control information, altered recirculation route, sensor uncertainty or unresolved distal-outlet coverage prevents a hygiene conclusion. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not certify water hygiene, declare legionella risk controlled/uncontrolled, recommend flushing or temperature changes, or infer outlet safety from plant telemetry. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-034 Do cold-water temperature patterns justify a competent-person water-hygiene investigation?**SENSORS AND TELEMETRY**

Proposed: Cold-water storage/branch temperatures at verified engineering monitoring points, ambient plantroom/riser temperature and flow evidence where useful. Coverage: Cold-water tanks and selected distribution monitoring points defined by the responsible water-hygiene regime; no fixture/user tracking. Timing: 1-5 min temperature trends summarized by duration and recurrence, plus competent-person spot checks at their actual test times. Limits: Fixed sensors may be poorly located or thermally coupled to room/pipe surface; branch temperature does not establish all outlets or stagnation. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis + recommendation. Decision: At the interval defined by the responsible person and when persistent abnormal temperature patterns appear. Minimum evidence: Current control criteria from the authoritative regime, verified sensor location/accuracy and repeated observations sufficient to distinguish transient ambient effects from persistent conditions. Method: Report duration/frequency of excursions at each monitoring point with ambient/flow context and compare with competent-person spot checks where available. Uncertainty: Telemetry cannot detect microbiological contamination, dead legs or every outlet temperature. Safe failure: Escalate persistent/conflicted patterns to the responsible person; do not assign hygiene risk or corrective action automatically.

WHY THIS IS USEFUL

The engineer can escalate credible temperature concerns to the responsible water-hygiene process without misrepresenting BMS trends as compliance evidence.

AUTHORITATIVE SUPPORTING DATA

Owners: Responsible person for water hygiene, current risk-assessment/control-regime owner, public-health engineer and competent water-hygiene service provider. Precedence: Current risk assessment/control scheme and competent-person measurements/inspection records take precedence, followed by current system drawings and verified telemetry as trend evidence. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Unknown sensor immersion/contact quality, old risk assessment or changed pipework means telemetry is a trigger only, not a control-status statement. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not diagnose legionella risk, certify compliance, instruct flushing/disinfection, change temperatures, or substitute telemetry for the competent water-hygiene control scheme. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-035 Is there enough evidence that water pressure and flow are adequate at the hydraulically difficult parts of the system?**SENSORS AND TELEMETRY**

Proposed: Verified pressure transducers at plant/branch points, booster pump speed/status and branch flow meters where fitted; temporary pressure/flow logging can be proposed for critical outlets by competent staff. Coverage: Supply plant, riser/branch and nominated hydraulically remote points relevant to the engineering requirement. Timing: 1-5 s to 1 min during peak draw-off tests where authorised; 5 min summaries over normal operation; temporary test data retains exact test configuration. Limits: Static pressure does not prove dynamic performance; unknown simultaneous draw-off, regulator settings and fixture restrictions affect results. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + forecast. Decision: During low-pressure complaints, change-of-use assessment or before adding significant demand. Minimum evidence: Current pressure/flow requirement, verified supply/branch mapping and dynamic measurements at representative high-demand conditions or a defined test. Method: Compare static and dynamic pressure, pump response and branch flow; identify pressure drop during demand and determine which remote points require direct competent measurement. Uncertainty: Aggregate telemetry may miss local restrictions and future simultaneous demand can exceed observed conditions. Safe failure: Return a measured operating envelope and required direct test; do not extrapolate unmeasured outlet performance.

AUTHORITATIVE SUPPORTING DATA

Owners: Public-health/mechanical engineer, Estates water-supply owner, commissioning/test owner and maintenance owner. Precedence: Current design pressure/flow requirements and system schematic, competent-person dynamic pressure/flow test results and current booster/regulator settings, then verified fixed telemetry. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: If test point identity, gauge calibration, booster configuration or current demand requirement is unknown, adequacy remains not verified. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not guarantee outlet pressure/flow, approve new demand, direct valve/regulator changes or infer pipe sizing/condition from pressure trend alone. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can determine whether the issue is supply/booster/branch related and specify the smallest dynamic test needed for design confidence.

ME-036 Is there engineering evidence of an emerging drainage blockage, surcharge or local flooding condition in a monitored area?**SENSORS AND TELEMETRY**

Proposed: Sump/chamber level where safely instrumented, high-level alarms, drainage pump status, local water-presence sensors and rainfall context where relevant. Coverage: Only monitored drainage assets/chambers and adjacent plant areas; no assumption about uninstrumented underground branches. Timing: Level/pump events at 1-60 s with 15 min rainfall/context summaries; preserve event rise/fall slopes and alarm duration. Limits: A level sensor can foul or drift; rainfall ingress, pump faults and downstream restriction can create similar level patterns; underground topology may be uncertain. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis. Decision: After repeated high-level events, abnormal pump cycling or before critical wet-weather periods/work. Minimum evidence: Verified monitored asset identity and level datum, current drainage topology for the affected section and repeated event context or direct inspection evidence. Method: Characterize level rise/recovery, pump response and rainfall dependence; compare similar events and rank local pump, downstream restriction or inflow hypotheses. Uncertainty: Telemetry cannot see uninstrumented blockages or prove pipe structural condition. Safe failure: Escalate abnormal patterns for competent inspection/CCTV/cleaning assessment; do not state a pipe is clear or blocked without direct evidence.

AUTHORITATIVE SUPPORTING DATA

Owners: Public-health/drainage engineer, Estates drainage asset owner, maintenance owner and current drainage drawing/survey owner. Precedence: Current drainage schematics/CCTV or survey records and competent-person inspection findings, then verified pump/level telemetry and rainfall context. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Changed drainage routes, inaccessible chambers, stale survey evidence or unverified level sensor datum keeps location/cause uncertain. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not direct entry into confined spaces, certify drainage condition, infer hidden pipe routes or authorize pumping/clearing operations. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can prioritize drainage inspection before a service failure and choose whether the next check should be pump, chamber or downstream survey.

ME-037 Is this sump or drainage pump cycling or running in a pattern that warrants maintenance investigation?**SENSORS AND TELEMETRY**

Proposed: Pump run status, start count, runtime, current/power where metered, sump level and duty/standby sequence states. Coverage: The nominated sump set, level sensors and discharge context where mapped. Timing: Native starts/stops with 1-60 s level/power; daily/weekly start-rate and runtime summaries with event-level traces. Limits: A status point may represent command rather than proven run; short cycling may result from level-switch spacing, inflow, non-return leakage or control faults. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis + recommendation. Decision: Weekly/monthly condition screening and after high-level or pump-fault events. Minimum evidence: Verified pump run feedback or power evidence, level signal, current sequence intent and enough events to establish recurring behavior. Method: Calculate cycle duration, starts/hour, level excursion and duty/standby balance; flag dry-running-like, short-cycle or non-response signatures for inspection. Uncertainty: Telemetry cannot determine impeller wear, blockage or check-valve condition without direct tests. Safe failure: Return a maintenance investigation trigger and evidence trace, not a component diagnosis.

AUTHORITATIVE SUPPORTING DATA

Owners: Drainage/mechanical engineer, maintenance owner, BMS/controls owner and latest pump service/test record owner. Precedence: Current pump/control design and duty/standby sequence, latest competent-person maintenance/functional test results, then verified telemetry. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Changed float/level settings, replaced pump, uncertain run feedback or missing discharge evidence makes root cause not verified. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not command pumps, alter level setpoints, declare a pump safe/failed from telemetry alone, or authorize confined-space access. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can schedule targeted pump/control/valve inspection before repeated cycling becomes a service or reliability problem.

ME-038 Do main and sub-water meters reconcile well enough to support branch-level engineering conclusions?**SENSORS AND TELEMETRY**

Proposed: Main and nested submeter volume/rate data with pulse multipliers, meter direction and reset/replacement events. Coverage: The complete meter tree for the selected water boundary, including known unmetered branches. Timing: 15 min and daily totals over at least several normal weeks; align meter clocks and quantify missing intervals before reconciliation. Limits: Meter accuracy classes differ; storage tanks, intermittent fill and timestamp boundaries can create short-term imbalance even when totals are valid. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis. Decision: Before branch baselining, leak localization or savings verification. Minimum evidence: Verified meter hierarchy/scaling and synchronized data across all major branches for a representative period. Method: Reconcile cumulative volumes over matching windows, propagate known meter tolerances, identify persistent residual by branch and distinguish storage/timing effects from structural mismatch. Uncertainty: A mathematically balanced tree can still have wrong physical mapping if errors cancel. Safe failure: If residual exceeds the evidence tolerance or mapping is uncertain, branch conclusions are NOT VERIFIED and meter checks are specified.

AUTHORITATIVE SUPPORTING DATA

Owners: Water utility/metering owner, public-health engineer, asset-information owner and meter maintenance owner. Precedence: Current meter hierarchy, installation/multiplier records and competent-person meter verification, then validated interval data; inferred meter nesting is not authoritative. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Any meter replacement, unknown pulse factor, duplicated/nested boundary or unexplained reversal is a conflict requiring resolution. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not invent consumption for unmetered branches, infer users from branch flow, or claim leak-free performance from meter balance alone. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can know whether submeter data is fit for leak, retrofit or operational analysis before relying on branch-level numbers.

ME-039 What is the engineering demand pattern on this water branch without inferring individual behaviour?**SENSORS AND TELEMETRY**

Proposed: Mapped branch flow/volume meter, storage/booster state and aggregate schedule/context data that is already authorised. Coverage: One engineering branch or system zone; no fixture-level attribution unless the meter is an approved engineering asset and no person can be identified. Timing: 1-15 min flow summarized by hour/day type, peak interval, minimum flow and event duration over representative weeks. Limits: Storage tanks and pressure controls decouple branch meter timing from actual downstream draw-off; low resolution can merge multiple events. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + estimate + forecast. Decision: For branch sizing checks, storage/booster studies and retrofit baselines. Minimum evidence: Verified branch boundary and enough data to capture normal/peak day types plus current engineering purpose/constraints of the branch. Method: Compute peak flow intervals, duration curves and day-type patterns; compare with booster/storage response and design assumptions without inferring who generated demand. Uncertainty: Observed peak may not represent design simultaneous demand and future change of use can invalidate the profile. Safe failure: Report the measured operating pattern and state separately what design-demand calculation remains required.

AUTHORITATIVE SUPPORTING DATA

Owners: Public-health/mechanical engineer, water/metering owner and current branch/service mapping owner. Precedence: Current meter-to-branch mapping and design/service information, then verified interval data and current operational schedules where relevant. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Changed branch use, meter boundary or unverified space/service assignment prevents interpretation beyond the measured aggregate pattern. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not infer occupant routines, research activity, hygiene behavior or future design flow solely from historical branch telemetry. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can compare real aggregate demand with design assumptions and decide whether temporary higher-resolution logging or a new design calculation is needed.

ME-040 Which services are documented as affected by a planned water isolation, and which dependencies remain unverified?**SENSORS AND TELEMETRY**

Proposed: Valve position feedback, branch pressure/flow and dependent plant make-up status may provide context during an authorised isolation only. Coverage: The isolation valve/section, known downstream branches, tanks, equipment make-up and any documented cross-connections. Timing: Current pre-work state and event telemetry during authorised operations; never initiate or schedule valve action through the service. Limits: Valve indication does not prove tight isolation; stored water, tanks, cross-feeds, bypasses and undocumented branches can sustain flow. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + calculation. Decision: During shutdown planning and immediately before on-site competent verification. Minimum evidence: Current isolation procedure/schematic, verified valve/branch identity and current dependency information for all potentially affected critical services. Method: Trace documented downstream dependencies, identify tanks/cross-feeds and classify services confirmed, possible or unknown; produce a field-verification and stakeholder-notification list. Uncertainty: A dependency graph cannot prove valve operability, tight shutoff or residual pressure. Safe failure: Return NOT VERIFIED for unresolved branches and defer execution to the applicable competent-person isolation process.

AUTHORITATIVE SUPPORTING DATA

Owners: Water/public-health responsible owner, maintenance/isolation procedure owner, mechanical engineer and affected service owners. Precedence: Current approved isolation schematic/procedure and current asset labels, then competent-person site verification and required tests; telemetry is supporting context only. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Unknown valve identity, stale schematic, unresolved bypass/cross-connection or critical dependency keeps the isolation impact not verified. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not state that water is isolated, safe to open, safe to drain or restored; do not operate valves or replace permit/shutdown procedures. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can plan outage impact and field verification while avoiding false confidence from a valve icon or historical schematic.

ME-041 Which BMS points are physically verified against the assets and sensors they claim to represent?**SENSORS AND TELEMETRY**

Proposed: BMS point names, object paths, live values, controller references and authorised field checks of sensor/actuator labels; temporary stimulus is used only within a competent-person test. Coverage: The selected plant, controller and engineering points needed for the decision, not the entire controls network by default. Timing: Current point inventory plus live-state snapshots and authorised test windows where applicable; record exact controller/point timestamps. Limits: A plausible value does not prove location; duplicated point names, virtual values, software scaling and replaced sensors can preserve misleading labels. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis + recommendation. Decision: Before diagnostics, recommissioning, analytics onboarding or controls modification design. Minimum evidence: Current point/I-O schedule plus a verified asset identity and direct physical/functional evidence for decision-critical points. Method: Match object path, engineering units, range and physical location; use authorised stimulus/response only where permitted; classify each point verified, probable, conflicted or unmapped. Uncertainty: Correlated response can occur because several points share a sequence or plant condition. Safe failure: Exclude unverified points from decisive analytics and return the exact field test needed to establish mapping.

AUTHORITATIVE SUPPORTING DATA

Owners: BMS/controls owner, controls engineer, asset-information owner and discipline engineer. Precedence: Current approved point schedule/I-O list and controls drawings, current physical labels and latest competent-person commissioning/test evidence, then live BMS values. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Unresolved controller migrations, renamed points, replaced sensors or inconsistent physical labels keep the point mapping unverified. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not write to controllers, initiate functional stimuli, infer hidden network topology, or present an unverified BMS label as physical truth. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can separate trustworthy BMS points from assumed mappings before using trends for diagnosis, commissioning or control-sequence assessment.

ME-042 Is the implemented BMS sequence behaving like the current approved control intent across relevant operating modes?**SENSORS AND TELEMETRY**

Proposed: Commands, feedbacks, setpoints, mode flags, temperatures/pressures, plant enables and interlock states required to observe the approved sequence. Coverage: The selected control loop/system and its upstream/downstream interlocks; safety systems are included only as read-only approved status evidence. Timing: 1-10 s for state transitions and 1-5 min for slow loops across normal start-up, occupied, setback and shutdown modes; use authorised functional tests where needed. Limits: Missing internal logic states, undocumented software changes and supervisory overrides can make external points insufficient to prove code behavior. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis. Decision: After software/plant changes, during recommissioning and when recurring mode faults occur. Minimum evidence: Current approved sequence, verified observable points for each material transition and at least one representative event in every decision-relevant mode. Method: Translate the approved sequence into observable states/conditions, compare event order and timing, and distinguish missing evidence from true deviation. Uncertainty: Not all logic is externally observable; permissives and safety interlocks may be intentionally hidden or unavailable. Safe failure: Return observed conformance/deviation only for observable clauses and require competent controls inspection/test for hidden logic.

AUTHORITATIVE SUPPORTING DATA

Owners: Controls/BMS owner, discipline engineer, controls contractor/commissioning record owner and authorised change-control owner. Precedence: Current approved sequence of operation/cause-and-effect and authorised controller configuration/version record, then competent-person functional test evidence and verified telemetry. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: If the approved sequence or current controller version is unknown, the service can describe observed behavior but cannot say it conforms to intent. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not modify code, force points, override interlocks, bypass safeties or approve a controls sequence as compliant. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can identify exactly which sequence clauses are behaving unexpectedly and give the controls team a reproducible evidence trace.

ME-043 Which BMS overrides, manual modes or forced values are active, how long have they persisted, and what functions do they affect?**SENSORS AND TELEMETRY**

Proposed: Read-only override/force/manual flags, commanded and actual values, operator annotations where authorised and dependent plant states. Coverage: Selected systems plus dependencies affected by the override; user identity is not exposed unless legitimately required by the authoritative operational system. Timing: Native event log and current-state snapshot; calculate duration from the authoritative event/start time and show gaps where history is unavailable. Limits: Some controllers do not expose every force; a manual hardware selector may not appear in BMS data, and stale UI flags can persist after controller changes. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + calculation + diagnosis. Decision: Daily/weekly operational assessment and before commissioning or fault diagnosis. Minimum evidence: Current verified override state, event history where available, affected point mapping and current control intent. Method: Classify active overrides by scope/duration, trace their downstream control effect and distinguish temporary authorised interventions from unexplained persistent states. Uncertainty: An override can be necessary for safe temporary operation; duration alone does not make it incorrect. Safe failure: Surface unresolved long-lived overrides to the authorised controls/maintenance owner; do not clear them automatically.

AUTHORITATIVE SUPPORTING DATA

Owners: BMS/controls operational owner, duty/maintenance owner and discipline engineer. Precedence: Current controller/operator override state and authorised operational record/change ticket, then current sequence/asset mapping and verified telemetry. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Overrides with no authoritative start/reason/owner, inconsistent controller state or uncertain point mapping are flagged unresolved rather than assigned a cause. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not cancel overrides, disclose operator identity unnecessarily, label an authorised temporary state unsafe without competent assessment, or change plant operation. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can prevent an old override from invalidating diagnostics and can prioritize which persistent manual states need owner confirmation or recommissioning.

ME-044 Are conflicting schedules, calendars or optimum-start rules causing plant to run outside approved service windows?**SENSORS AND TELEMETRY**

Proposed: BMS occupancy/enable schedules, optimum-start/stop state, plant enable/run feedback and aggregate energy/temperature context. Coverage: The selected plant and engineering zones with current authoritative service schedules; no inference of room occupancy from plant running. Timing: Native schedule changes plus 1-5 min enable/run states over at least two normal weeks including weekends/holidays where relevant. Limits: A plant may run for frost protection, hygiene, process or safety reasons outside nominal hours; schedule labels may be inherited or overridden. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation. Decision: Monthly controls assessment, after timetable/service changes and when out-of-hours operation rises. Minimum evidence: Current approved operating windows, verified controller schedules/calendars and run/enable feedback for representative weeks. Method: Compare requested versus actual run windows, identify overlapping schedule sources and classify each out-of-window episode by explicit control reason where observable. Uncertainty: Weather compensation, optimum start and protective modes can legitimately extend schedules; missing mode flags reduce certainty. Safe failure: Report unexplained schedule conflicts and require owner confirmation before any change.

AUTHORITATIVE SUPPORTING DATA

Owners: Operational/BMS owner, current space/service-schedule owner, discipline engineer and approved controls change owner. Precedence: Current authoritative service/operating requirement and approved schedule/sequence, then current controller calendars and verified plant telemetry. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: If the current service window is unknown, telemetry can identify run periods but cannot call them early/late or unnecessary. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not infer actual occupancy, shorten schedules, disable protective modes or change calendars/setpoints automatically. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can remove contradictory schedule logic through controlled recommissioning and target energy/runtime problems without compromising required service windows.

ME-045 Which BMS sensors show credible evidence of bias, drift, sticking or implausible dynamics that warrants calibration testing?**SENSORS AND TELEMETRY**

Proposed: The suspect sensor value, redundant/reference sensors where legitimately comparable, command/plant context and portable calibrated measurements when taken by competent staff. Coverage: Decision-critical temperature, pressure, humidity, CO₂, flow and electrical points with known locations; peer sensors are compared only when physical conditions are comparable. Timing: Native 10 s-5 min trends over weeks/months for drift plus exact timestamps for portable reference tests. Limits: Peer disagreement can reflect real spatial gradients; a stable signal may be correct; portable instruments have their own uncertainty and response time. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis + recommendation. Decision: Before decisions that depend on the point, during planned calibration cycles and when control behavior becomes implausible. Minimum evidence: Verified sensor identity/location, applicable accuracy requirement and either a trustworthy reference measurement or repeated physical-consistency evidence. Method: Check range/stuck values, rate-of-change, cross-variable physical constraints and reference offset; separate evidence of drift from general disagreement and rank by decision impact. Uncertainty: Analytics can identify suspicious behavior but cannot establish metrological traceability or exact calibration error without a valid reference. Safe failure: Mark the point SUSPECT and exclude it from decisive calculations until competent calibration/test evidence is available.

WHY THIS IS USEFUL

The engineer can prioritize calibration effort on sensors whose uncertainty materially affects control, energy or diagnostic decisions.

AUTHORITATIVE SUPPORTING DATA

Owners: BMS/controls owner, calibration/test record owner, discipline engineer and maintenance owner. Precedence: Current calibration/test certificates and competent-person reference measurements at the same location/condition, then approved sensor specification and verified telemetry history. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Expired calibration, changed sensor location/range, unknown scaling or non-comparable reference conditions prevents a bias value being asserted. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not auto-correct sensor values, adjust calibration offsets in the BMS, or certify calibration from analytical comparison. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-046 Are meter identities, units, CT ratios, pulse multipliers and scaling configured consistently enough for engineering use?**SENSORS AND TELEMETRY**

Proposed: Raw and engineering-unit meter registers, instantaneous voltage/current/power, pulse counts, BMS scaled values and known test loads where observed during authorised testing. Coverage: Selected electrical, thermal and water meters plus their gateway/BMS representations. Timing: Current configuration snapshot and synchronized 1-15 min data; compare cumulative and instantaneous channels over defined verification windows. Limits: Incorrect CT ratio or pulse multiplier can produce plausible but wrong data; gateway rescaling can duplicate scaling; sign convention and unit conversion errors can cancel in totals. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + estimate. Decision: Before energy/water baselining, savings verification or any capacity calculation using the meter. Minimum evidence: Verified physical meter identity, current scaling/configuration records and at least one independent plausibility check against raw registers or competent test. Method: Trace the conversion chain from raw register/pulse to final engineering unit, compare cumulative versus interval totals and detect unit/order-of-magnitude discrepancies. Uncertainty: Plausibility alone cannot prove scaling when the true load is unknown. Safe failure: Quarantine disputed meter channels from quantitative conclusions and specify the field/configuration check required.

WHY THIS IS USEFUL

The engineer can stop a single scaling error from contaminating energy, capacity and retrofit decisions across multiple systems.

AUTHORITATIVE SUPPORTING DATA

Owners: Metering/energy owner, electrical/mechanical engineer, BMS integration owner and meter commissioning/test record owner. Precedence: Current meter installation/configuration records and competent-person commissioning checks, then manufacturer meter registers and approved integration scaling; analytics comparison is supporting evidence. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Unknown CT/pulse factor, meter replacement, firmware/gateway change or disagreement between raw and scaled registers is treated as a configuration conflict. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not change meter/gateway scaling, write CT ratios, certify billing accuracy or use a disputed channel in savings/capacity claims. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-047 How healthy and complete is the read-only integration path from BMS and meters for this engineering decision?**SENSORS AND TELEMETRY**

Proposed: Point freshness, sample cadence, missingness, duplicate timestamps, value ranges, gateway status and source-side comparison for a defined set of engineering points. Coverage: Only the points needed for the engineering question, from physical source through gateway/API to the service. Timing: Continuous health metrics with daily summaries; decision windows use observed timestamps and report retrieval latency separately. Limits: A live API can still deliver stale cached data; missingness can be systematic during outages; resampling can hide fast events. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + forecast. Decision: Before every high-consequence analysis and continuously for operational data quality. Minimum evidence: Known source point IDs, expected cadence/units, source-side comparison and permissioned lineage through each transformation. Method: Calculate freshness, completeness, latency and transformation checks; compare source and ingested values on sampled windows and label each point fit, degraded or unusable for the decision. Uncertainty: Transport health does not prove sensor accuracy or physical mapping. Safe failure: Exclude degraded/unverified channels and return a data-quality limitation rather than filling gaps invisibly.

AUTHORITATIVE SUPPORTING DATA

Owners: BMS/metering integration owner, source-system owner, cybersecurity/access owner and discipline engineer using the data. Precedence: Source-system values and authoritative integration/configuration records, then verified transport/gateway logs and service-ingested data; the service copy never outranks the source. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Any permission loss, mapping change, unexplained lag, resampling or source/service disagreement is surfaced with affected dates/points. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not bypass access controls, inspect unrelated network content, write back to source systems or imply that integration health proves physical sensor validity. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can know whether an apparent plant anomaly is real or simply an integration/data-pipeline problem before acting on it.

ME-048 Where do command and feedback states persistently disagree for valves, dampers, fans, pumps or electrical indications?**SENSORS AND TELEMETRY**

Proposed: Command, position/run feedback, end-switch status, actuator output and relevant process response such as flow, pressure, temperature or current. Coverage: Selected actuated devices and the process measurement needed to interpret movement; safety-related switching remains read-only. Timing: 1-10 s around commands and 1 min steady-state summaries; calculate response delay and persistent mismatch after a device-appropriate settling time. Limits: A command-feedback mismatch can come from point mapping, actuator travel, hand/off/auto state, interlock, feedback contact or true mechanical/electrical failure. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + calculation + estimate + diagnosis + recommendation. Decision: Daily/weekly fault screening and during functional diagnosis. Minimum evidence: Verified command and independent feedback points, current intended sequence and enough repeated events to distinguish transient travel from persistent mismatch. Method: Measure response delay, mismatch duration and process consequence; separate no-command, no-feedback and no-process-response cases and rank likely information/control/device causes. Uncertainty: Telemetry cannot determine linkage condition, valve leakage, motor contact condition or actuator torque without testing. Safe failure: Raise a targeted functional-test recommendation; do not label the component failed until competent evidence confirms it.

AUTHORITATIVE SUPPORTING DATA

Owners: Controls/BMS owner, discipline engineer, maintenance owner and latest functional-test/commissioning record owner. Precedence: Current sequence, device specification and competent-person functional test evidence, then verified command/feedback/process telemetry. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Unknown feedback semantics, device replacement, manual selector state or unverified interlock keeps diagnosis open. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not force commands, cycle devices, bypass interlocks, operate switching devices or certify actuator/device function from trends alone. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can separate controls/mapping issues from likely field-device problems and give maintenance a precise test condition.

ME-049 Are timestamps across BMS, meters and portable instruments aligned closely enough for event diagnosis?**SENSORS AND TELEMETRY**

Proposed: Source timestamps, server/gateway receive times, known clock-sync status and simultaneous physical event markers visible across systems. Coverage: The specific sources used in the diagnostic timeline; no need to collect unrelated platform logs. Timing: At each test/incident, retain raw timestamps and synchronization offsets; use repeated common events to estimate drift over the window. Limits: Systems can use local/UTC time, daylight-saving changes, buffered uploads and coarse logging; apparent ordering may invert when offsets exceed event separation. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + calculation + estimate + diagnosis. Decision: Before cross-system event causality analysis and for every commissioning test using multiple loggers. Minimum evidence: At least one common event or trusted time reference for each source, plus raw timestamps and device clock configuration. Method: Estimate offset/drift, convert to a common time basis, attach earliest/latest bounds to events and only infer order where bounds do not overlap. Uncertainty: A clock-corrected sequence still does not prove causation. Safe failure: If offsets cannot be bounded tightly enough, present events as unordered/overlapping and specify a synchronized retest.

AUTHORITATIVE SUPPORTING DATA

Owners: BMS/integration owner, metering owner, test-instrument custodian and discipline engineer. Precedence: Authoritative time configuration and instrument settings, then known synchronized event records and raw source timestamps; service receipt time is not substituted for observation time. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Unknown timezone, DST handling, rebooted clocks or drifting portable instruments are explicit timing uncertainties. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not claim causal order from receipt timestamps, alter source clocks remotely, or hide timing uncertainty in merged plots. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can determine whether multi-system event correlation is credible or whether clock correction and repeat testing are needed first.

ME-050 Can BMS, meter, asset and maintenance data be joined without creating false physical relationships?**SENSORS AND TELEMETRY**

Proposed: Source-system identifiers, asset tags, point/meter parent references, location codes and time-series behavior used only as secondary corroboration. Coverage: The minimum entities needed for the engineering question; joins stop at authoritative asset/system boundaries. Timing: Current metadata snapshot with effective dates for mappings and time series only after identity confidence is established. Limits: Names and locations are often non-unique; historic maintenance records may use retired tags; correlated operation is not proof of common asset identity. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + calculation + forecast + diagnosis + recommendation. Decision: Before any multi-source fault, maintenance, energy or lifecycle analysis. Minimum evidence: At least one authoritative common identifier or owner-approved mapping for each decisive join, plus effective dates that cover the analysis period. Method: Construct a temporal relationship graph, prevent joins outside validity intervals, detect many-to-many collisions and label candidate joins rather than forcing them. Uncertainty: Assets can move or be replaced while software IDs remain constant, so identity may change over time. Safe failure: Exclude candidate/ambiguous joins from causal or quantitative outputs and return a mapping-resolution task.

AUTHORITATIVE SUPPORTING DATA

Owners: Asset-information owner, CMMS owner, BMS owner, metering owner and discipline engineer. Precedence: Current master asset IDs and approved cross-system mapping tables, then controlled drawings/current physical tags and verified system metadata; analytical similarity is lowest precedence. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: One-to-many mappings, reused tags, location conflicts or mapping tables with unknown effective dates remain unresolved. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not join records solely by similar names, location text or time-series correlation, and do not write inferred relationships back to authoritative systems. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can combine maintenance, controls and metering evidence without accidentally attributing a fault, runtime or energy value to the wrong asset.

ME-051 For this recurring fault, what repeatable evidence signature exists without deliberately provoking unsafe or disruptive operation?**SENSORS AND TELEMETRY**

Proposed: Read-only fault/alarm events, commands/feedback, process values, power and relevant environmental variables around naturally occurring or authorised test events. Coverage: The affected asset/system and directly relevant upstream/downstream context; no unrelated user or network content. Timing: Highest available event resolution with a defined pre/post window for multiple independent occurrences; keep raw events separate from resampled trends. Limits: Alarm duplication, clock skew and missing data during failures can distort signatures; repeated correlation does not by itself prove root cause. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis + recommendation. Decision: After at least two comparable natural/authorised occurrences and before intrusive diagnostic testing. Minimum evidence: Verified asset mapping, time-aligned event data for multiple occurrences and at least one non-fault comparison period. Method: Align on the first independently observed fault marker, identify signals consistently leading/following it, measure variability and test whether the pattern also appears without the fault. Uncertainty: A repeatable precursor may be a symptom of an upstream cause or common operating condition. Safe failure: Return the repeatable signature and discriminating next test; never trigger a fault or bypass safeguards to collect data.

WHY THIS IS USEFUL

The engineer can design a focused diagnostic test around a reproducible signature and avoid unsafe trial-and-error fault provocation.

AUTHORITATIVE SUPPORTING DATA

Owners: Discipline engineer, maintenance/fault record owner, BMS/controls owner and competent test owner. Precedence: Current fault/work-order evidence and competent-person findings, current sequence/design information and verified event telemetry; anecdotal reports are contextual unless time-anchored. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Changed configuration between events, unresolved timestamps or different failure modes are segmented rather than pooled. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not deliberately induce trips, disable safeties, cycle equipment outside authorised testing or declare root cause from pattern recurrence alone. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-052 During a competent-person functional test, did observable plant behaviour follow the approved test script and control intent?**SENSORS AND TELEMETRY**

Proposed: All read-only states/measurements explicitly required by the test: commands, feedbacks, interlocks, setpoints, temperatures/pressures, power and alarms. Coverage: Only assets and points within the approved test boundary; the competent tester controls all physical actions. Timing: 1-10 s or faster where transition timing matters, starting before the first authorised action and ending after stable recovery; preserve native event logs. Limits: The service may not see hidden PLC logic, physical actuator travel or safety-device state and cannot validate unobservable test steps. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + forecast + recommendation. Decision: For each authorised functional performance test or recommissioning test. Minimum evidence: Approved test script, verified measurement set, synchronized timestamps and competent-person record of actual test actions. Method: Map each test step to observable evidence, compare expected/observed state transition and timing, and label supported, deviated or unobservable without replacing tester judgement. Uncertainty: Telemetry can confirm only observable consequences, not physical safety checks or hidden logic. Safe failure: Produce a trace and evidence gaps; leave acceptance/pass-fail to the competent test authority.

WHY THIS IS USEFUL

The engineer can obtain a reproducible evidence trail for functional testing and identify exactly which test steps need direct observation or additional instrumentation.

AUTHORITATIVE SUPPORTING DATA

Owners: Commissioning authority/test lead, discipline engineer, BMS/controls owner and approved test-document owner. Precedence: Current approved test script/sequence and competent-person observations/results, then verified telemetry as time-stamped evidence; the tester determines pass/fail under the applicable process. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Any test deviation, missing point, unverified mapping or clock uncertainty is recorded and prevents automated pass/fail for that step. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not initiate test actions, force points, bypass interlocks, issue commissioning acceptance or certify the system. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

Diagnostics, testing, commissioning and evidence quality

ME-053 Which sensors should be calibrated or replaced first because their uncertainty most affects current engineering decisions?

SENSORS AND TELEMETRY

Proposed: Sensor histories, reference/portable measurements, redundant physical relationships and the calculations/control loops that consume each point. Coverage: Decision-critical sensors in the selected systems; rank by consequence, not by number of alarms. Timing: Long-term drift trends plus exact timestamps of competent reference checks; use recent representative operating data to estimate decision sensitivity. Limits: A noisy or offset point may have little decision impact, while a small bias in a flow/energy-balance sensor can materially change conclusions. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + calculation + estimate + diagnosis + recommendation. Decision: During calibration planning and before high-consequence diagnostics/verification. Minimum evidence: Current sensor inventory, decision/use mapping, applicable accuracy need and evidence of suspected error or uncertainty for candidate points. Method: Estimate how plausible sensor error changes the engineering decision/control outcome, combine with evidence of drift/sticking and rank calibration actions by decision consequence. Uncertainty: Sensitivity estimates depend on the model/decision context and cannot quantify unknown systematic errors reliably. Safe failure: Mark high-impact uncertain sensors as unavailable for decisive analysis until competent calibration/reference testing is completed.

AUTHORITATIVE SUPPORTING DATA

Owners: Calibration/test record owner, discipline engineer, BMS owner and maintenance planner. Precedence: Competent-person calibration/reference evidence and applicable sensor accuracy requirement, then current control/design use of the point and verified telemetry behavior. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Unknown calibration status, sensor replacement or changed use in calculations is explicit; analytics does not assign a calibration certificate. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not certify calibration, write offset corrections, extend calibration intervals or replace maintenance metrology requirements. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can spend calibration effort where measurement uncertainty most threatens control, capacity, energy or fault conclusions.

ME-054 What is the most defensible event timeline for this intermittent M&E fault across power, controls and plant states?

SENSORS AND TELEMETRY

Proposed: Synchronized electrical events, BMS alarms/states, motor/valve feedback, temperatures/pressures and relevant controller restart indicators. Coverage: The affected service path and immediate dependencies only; collect no unrelated communications or user content. Timing: Highest native event resolution, with at least 15 min pre-fault and post-recovery context; retain clock uncertainty for every source. Limits: Buffering, differing sample rates and missing data can create artificial order; failure can stop the very data stream needed to observe it. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + calculation + diagnosis + recommendation. Decision: For each significant intermittent incident after data preservation. Minimum evidence: Verified mappings, clock-bounded source events and at least one authoritative physical/operational incident marker. Method: Normalize times, build a fact-only sequence with uncertainty intervals, then overlay alternative causal hypotheses and identify evidence that discriminates them. Uncertainty: Temporal precedence is necessary but not sufficient for causation and intermittent faults may have multiple triggers. Safe failure: Return the evidence timeline, confidence and next discriminating test rather than a single unsupported root cause.

AUTHORITATIVE SUPPORTING DATA

Owners: Discipline engineer, electrical/controls owners, maintenance incident owner and competent test/inspection owners. Precedence: Competent-person incident/switching/test records and current system topology/sequence, then verified event telemetry; user reports provide time/context only where permissioned. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Configuration changes, unresolved clocks or asset identity conflicts between incidents require separate timelines. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not claim causal certainty from event order, reset plant, change protection/control logic or expose unrelated operational/security data. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can coordinate electrical and controls troubleshooting around one evidence timeline instead of conflicting logs and recollections.

Diagnostics, testing, commissioning and evidence quality

ME-055 Which points and sampling rates should be trended so this commissioning or diagnostic test can observe the expected behaviour?

SENSORS AND TELEMETRY

Proposed: A proposed temporary/fixed trend set selected from the physical variables and states required by the test hypothesis, including appropriate calibrated portable instruments where necessary. Coverage: Only test-boundary assets plus the minimum upstream/downstream references needed to discriminate causes. Timing: Sampling is matched to system dynamics: sub-second/seconds for switching/fast controls where required, 10-60 s for actuators, and 1-5 min for slow thermal behavior; retain raw data. Limits: Oversampling cannot compensate for the wrong measurement location or inaccurate sensor; BMS trend capacity may downsample or drop points. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis + recommendation. Decision: Before every planned commissioning/diagnostic test. Minimum evidence: Explicit test hypothesis/acceptance variables, expected time constants, current point capability and required accuracy/location. Method: Map each hypothesis to direct measurements, state transitions and reference conditions; choose a sample interval at least several times faster than the expected material transition and document aggregation separately. Uncertainty: BMS storage, network latency and portable logger limits can constrain feasible cadence. Safe failure: If a decisive variable is unmeasured or too slow/inaccurate, state that the test is not observable and require appropriate instrumentation.

AUTHORITATIVE SUPPORTING DATA

Owners: Commissioning/test lead, discipline engineer, controls/BMS owner and instrument custodian. Precedence: Approved test objective/script and current design/control intent, then instrument capabilities/calibration and authoritative point mappings; existing trend defaults are not automatically adequate. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: If required transition time/measurement accuracy is unknown, the trend plan remains provisional and the test lead decides the final setup. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not define statutory acceptance criteria, install instruments unsafely, command plant or imply that a dense trend set substitutes for competent test procedure. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can collect enough evidence to answer the test question once, avoiding a failed commissioning exercise caused by missing or over-aggregated trends.

ME-056 How long, and under which operating conditions, must this variable be measured before the engineering conclusion is credible?

SENSORS AND TELEMETRY

Proposed: The decision variable plus drivers that explain variability, such as outdoor condition, load, plant mode, schedule, flow or power. Coverage: The defined system boundary and representative operating modes relevant to the decision. Timing: Duration is chosen from observed variability and decision interval: event tests may need minutes/hours; operational baselines may need representative weeks/seasons; preserve native data and summarize explicitly. Limits: A long dataset can still be unrepresentative if it misses peak load, hot/cold weather, changeover, weekends or abnormal modes. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + estimate + forecast. Decision: When planning temporary monitoring, baselines, performance tests or intervention verification. Minimum evidence: Defined decision metric/tolerance, expected variability drivers and preliminary measurements sufficient to estimate recurrence/variance. Method: Assess convergence of the metric, coverage of required operating modes and sensitivity to excluding days/events; stop only when predefined evidence conditions are met, not at an arbitrary date. Uncertainty: Rare peaks and seasonal conditions may remain unobserved, so confidence can plateau before true extremes are covered. Safe failure: State which operating modes remain untested and whether the conclusion is indicative or requires extended/seasonal monitoring.

AUTHORITATIVE SUPPORTING DATA

Owners: Discipline engineer, commissioning/test owner, operational owner and relevant design/measurement record owners. Precedence: Current engineering decision criteria/test objective and known operational/calendar/weather context, then verified measurements; generic fixed monitoring durations are secondary. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: If operating regime changed during monitoring or the required decision condition was never observed, duration alone does not make evidence sufficient. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not claim statistical duration alone establishes engineering adequacy, ignore unobserved design conditions, or substitute monitoring for required commissioning/statutory tests. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can choose a monitoring campaign that is long enough to be decision-useful without collecting months of low-value data.

ME-057 After remedial work, is evidence sufficient for technical acceptance consideration, or is targeted retesting still needed?**SENSORS AND TELEMETRY**

Proposed: Pre/post fault signature, relevant process values, commands/feedback, power, alarms and competent test measurements required by the approved acceptance plan. Coverage: The corrected asset/system and interfaces affected by the intervention. Timing: Use a defined post-work test window plus representative operation long enough to show recurrence risk where appropriate; keep before/after conditions comparable. Limits: Absence of an alarm over a short period is weak evidence; environmental/load changes can make post-work performance look better without proving the intervention caused it. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis + recommendation. Decision: Immediately after work and after a defined representative proving period where the acceptance plan requires it. Minimum evidence: Approved scope/criteria, competent test results, confirmed defect closure and enough post-work telemetry to exercise the original failure condition if safely/authoritatively observable. Method: Compare pre/post signatures under matched conditions, verify each acceptance criterion has direct evidence and identify criteria that remain untested. Uncertainty: Non-recurrence is probabilistic for intermittent faults and may require longer observation. Safe failure: Return evidence COMPLETE / INCOMPLETE with missing test steps; the appointed engineer/authority makes acceptance decisions.

AUTHORITATIVE SUPPORTING DATA

Owners: Discipline engineer/acceptance authority, maintenance/contractor completion owner, commissioning/test lead and BMS owner. Precedence: Approved remedial scope and acceptance/test criteria plus competent-person test/completion evidence, then verified operational telemetry as supplementary confirmation. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Incomplete completion records, changed test conditions, missing acceptance criteria or unresolved defects keep acceptance not verified. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not issue completion/acceptance certification, close defects automatically, or treat absence of alarms as proof of successful rectification. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can see whether the remedial package has objective evidence behind it and request only the missing retests before closeout.

ME-058 Did a recent plant or controls intervention improve performance after accounting for weather, load and operating-condition differences?**SENSORS AND TELEMETRY**

Proposed: Relevant energy/power, temperatures, pressures/flows, runtime and control states plus outdoor-condition/load proxies appropriate to the system. Coverage: The changed system and an unaffected reference system/period where physically comparable and permissioned. Timing: Pre/post periods long enough to include similar operating modes; 1-15 min source data aggregated to daily/condition bins for normalization while retaining raw traces. Limits: Concurrent maintenance, schedule/use changes and sensor recalibration can confound effects; a simple before/after kWh comparison is not causal. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + estimate + forecast. Decision: After a sufficient post-intervention proving period and before claiming sustained benefit. Minimum evidence: Confirmed intervention date/scope, valid pre/post measurements, service-performance variables and enough overlapping weather/load conditions for comparison. Method: Use matched-condition bins or a transparent baseline model, test residual change and confirm service indicators did not degrade; provide uncertainty/sensitivity to model choice. Uncertainty: Observational data may not isolate the intervention from all simultaneous changes. Safe failure: State an observed/estimated effect range and confounders; do not claim causality when counterfactual evidence is weak.

AUTHORITATIVE SUPPORTING DATA

Owners: Discipline engineer, intervention/change owner, energy/BMS owner and current operating-requirement owner. Precedence: Approved intervention scope/effective date and current service criteria, then verified meter/BMS data and competent functional evidence; modeled counterfactuals are explicitly analytical. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Unknown implementation date, concurrent changes or altered meter mapping are confounders that must be disclosed. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not guarantee savings, approve permanent control changes, hide service deterioration or present a modeled effect as a certified measurement-and-verification result. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can decide whether to retain, refine or further test an intervention based on comparable performance rather than raw before/after totals.

Diagnostics, testing, commissioning and evidence quality

ME-059 When a calibrated portable instrument and BMS disagree, is the likely issue bias, location, timing, scaling or true system variability?

SENSORS AND TELEMETRY

Proposed: BMS point, calibrated portable/reference instrument, exact measurement location, sampling cadence, response-time metadata and nearby process context. Coverage: The same physical quantity at the same or demonstrably comparable location and operating condition. Timing: Co-located simultaneous readings over enough time to cover sensor response and stable/transient conditions; retain raw values and instrument timestamps. Limits: A portable probe at a different height/immersion/depth may legitimately differ; thermal lag, averaging, radiant effects and logger clock offsets can create apparent bias. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + estimate + diagnosis. Decision: During calibration checks, commissioning or disputed diagnostic measurements. Minimum evidence: Traceable portable instrument, verified BMS sensor identity/location/scaling, simultaneous co-located data and stable conditions or an explicit dynamic-response test. Method: Align clocks, compare steady offset and dynamic response, account for stated instrument uncertainty and test whether disagreement follows location/process gradients. Uncertainty: A single short comparison cannot separate intermittent bias from real local variability. Safe failure: Report whether disagreement is statistically/engineering-significant and specify retest or calibration; do not declare either instrument correct without the evidence chain.

AUTHORITATIVE SUPPORTING DATA

Owners: Instrument custodian/calibration owner, BMS owner, discipline engineer and competent tester. Precedence: Valid calibration/reference certificate and competent-person test setup, then current BMS sensor specification/scaling/location and telemetry. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Expired reference calibration, uncertain co-location, changed scaling or unstable process conditions prevent assigning error to either instrument. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not overwrite BMS calibration, certify the fixed sensor, or choose a reading simply because the portable instrument is newer. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can resolve measurement disputes methodically and prevent a bad reference setup from driving unnecessary sensor replacement.

ME-060 Which current engineering decisions remain unanswerable because instrumentation, mapping, calibration, history or authoritative records are insufficient?

SENSORS AND TELEMETRY

Proposed: Metadata for available sensors/meters, physical mappings, calibration status, data completeness/freshness and temporary measurement capabilities; missing variables remain explicitly missing. Coverage: The exact systems and decisions in scope rather than a generic building-wide sensor inventory. Timing: Current evidence inventory plus relevant historical data availability; summarize direct, proxy, stale, conflicted and absent evidence by decision variable. Limits: Sensor presence does not mean correct location/accuracy/cadence; a proxy cannot silently become a direct measurement and more data cannot fix unknown system topology. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + estimate + diagnosis. Decision: At project/test mobilisation and before any analytics-dependent engineering conclusion. Minimum evidence: A defined decision, required variables/tolerances/time horizon and complete inventory of available authoritative and measured evidence. Method: Map each decision claim to direct measurement/record, proxy or model assumption; score coverage, accuracy, freshness and provenance; rank additional evidence by expected decision value. Uncertainty: Some questions remain unanswerable even after new sensors if concealed topology, legal authority or competent physical testing is the limiting factor. Safe failure: Return NOT VERIFIED and the smallest safe evidence-acquisition step rather than filling gaps with confident estimates.

AUTHORITATIVE SUPPORTING DATA

Owners: Discipline engineer, asset-information owner, BMS/metering owners, commissioning/test owners and current engineering record owners. Precedence: The decision requirement and current authoritative drawings/records define what must be known; calibrated tests and verified mappings follow; telemetry is used only when its evidence characteristics meet the requirement. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Any source outside its valid/effective period, unpermissioned source or unresolved conflict is unavailable for the decision even if technically accessible. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not fabricate missing evidence, infer concealed conditions, present model outputs as observations or imply that analytics can replace mandated inspection/testing. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can stop unsupported analysis early and specify the smallest additional instrumentation, record or test needed to make the decision answerable.

ME-061 Which M&E assets are due or overdue for maintenance when approved regimes and actual runtime or starts are both considered?**SENSORS AND TELEMETRY**

Proposed: Verified runtime, starts, load hours, alarm/fault counts and condition indicators only where the maintenance regime explicitly allows such evidence to inform timing. Coverage: The selected maintained asset classes with authoritative asset IDs and current maintenance strategy. Timing: Daily runtime/start accumulation with monthly summaries; preserve counters across controller resets/replacements and compare with CMMS completion dates. Limits: Controller runtime can reset, count commanded rather than proven operation, or omit off-BMS operation; high use does not automatically shorten every maintenance interval. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + recommendation. Decision: Weekly/monthly planning and before shutdown/mobilisation. Minimum evidence: Current maintenance rule for each asset, authoritative last-completion evidence, verified asset identity and valid runtime/start data where usage-based logic is permitted. Method: Calculate calendar-based due status first, then compare usage thresholds and identify assets where runtime evidence advances or deprioritizes discretionary maintenance within the approved regime. Uncertainty: Telemetry cannot determine whether a completed task was adequate, and statutory/manufacture rules may not permit extension based on low use. Safe failure: Where records conflict or mandatory intervals apply, use the conservative authoritative due status and refer to the maintenance owner.

WHY THIS IS USEFUL

The engineer can plan maintenance around genuine asset use while preserving mandatory regimes and focusing verification on questionable runtime or completion records.

AUTHORITATIVE SUPPORTING DATA

Owners: Maintenance/CMMS owner, discipline engineer, statutory/responsible owner where relevant and asset-information owner. Precedence: Current approved maintenance regime/manufacture requirements and statutory schedule where applicable, then completed competent-person maintenance records; verified telemetry can refine usage context but cannot override mandatory intervals. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Missing completion evidence, reset counters, asset replacement or conflicts between CMMS and physical asset identity are reported before due-date calculation. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not extend statutory/mandatory maintenance intervals, mark work complete from telemetry, or automatically reschedule CMMS tasks. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-062 Which assets are credible candidates for condition-based maintenance, and which still require time-based or statutory intervention?**SENSORS AND TELEMETRY**

Proposed: Asset-specific condition evidence such as vibration, bearing temperature, motor current/power, differential pressure, runtime, start count and fault signatures where those measurements are technically valid. Coverage: Rotating/mechanical/electrical assets for which failure modes and sensing coverage are explicitly defined; exclude assets whose critical degradation is not observable. Timing: Condition variables at a cadence appropriate to the failure mode, with long-term baselines and event capture where needed; summarize trend rate and operating-condition normalization. Limits: A stable proxy may miss sudden or hidden failure modes; sensor installation quality and changing load can dominate apparent trends. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + recommendation. Decision: During maintenance-strategy design and annually thereafter. Minimum evidence: Current maintenance obligations, credible failure modes, asset criticality and evidence that available measurements detect material degradation with adequate lead time. Method: Map each failure mode to direct/proxy condition evidence, estimate detection lead time and false-negative consequence, and separate tasks that can be condition-informed from tasks that remain fixed/mandatory. Uncertainty: Historical data may contain too few failures to validate thresholds, especially for critical assets. Safe failure: Use CBM only as an engineering recommendation for owner approval; retain fixed intervention where evidence is weak or obligations require it.

WHY THIS IS USEFUL

The engineer can identify where condition monitoring can add value without weakening maintenance duties or applying CBM to unobservable failure modes.

AUTHORITATIVE SUPPORTING DATA

Owners: Maintenance strategy owner, discipline engineer, statutory duty holder/responsible person where relevant and OEM/service record owner. Precedence: Mandatory/statutory and OEM maintenance requirements plus current failure-mode/criticality information take precedence; competent inspection/test evidence follows; telemetry supports only approved condition-based decisions. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: If failure mode, sensor-to-failure relationship, calibration or consequence is uncertain, the asset is not treated as a CBM candidate. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not cancel statutory/OEM tasks, predict exact failure dates, or claim a condition score certifies fitness for service. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

Maintenance, statutory duties and technical risk

ME-063 Which recurring alarms are distinct technical problems, and which are downstream symptoms, duplicates or configuration noise?

SENSORS AND TELEMETRY

Proposed: BMS/alarm event history with associated process values, commands/feedback, equipment states and current alarm configuration. Coverage: The selected plant/system and its dependent alarm tree; safety-critical alarms remain clearly separated from optimization candidates. Timing: Native event timestamps with 1-5 min context; summarize episode frequency, persistence, burst size and first/last alarm order over representative months. Limits: One physical event can produce many alarms; configuration changes and clock issues distort history; a low-frequency alarm may still be high consequence. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis + recommendation. Decision: Weekly/monthly alarm assessment and after significant controls changes. Minimum evidence: Current alarm configuration, verified point dependencies and enough event context to group repeated alarms into episodes. Method: Cluster temporally/dependently related alarms, identify likely initiating versus consequential signals, compare with confirmed fault outcomes and rank by consequence/persistence rather than count alone. Uncertainty: Historical association cannot prove an alarm is harmless or redundant, and rare safety alarms require separate governance. Safe failure: Recommend candidates for authorised rationalisation/testing only; preserve safety-critical alarms and unresolved signals.

WHY THIS IS USEFUL

The engineer can focus fault investigation on initiating signals and give the controls team evidence for disciplined alarm rationalisation rather than chasing alarm volume.

AUTHORITATIVE SUPPORTING DATA

Owners: BMS/alarm management owner, maintenance owner, discipline engineer and relevant safety/statutory responsible owner. Precedence: Current alarm philosophy/priority and approved control sequence, competent fault/maintenance evidence and authorised configuration records, then verified event telemetry. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Alarms lacking current priority rationale, point mapping or stable configuration are not labelled nuisance by analytics. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not suppress, inhibit, reprioritize or disable alarms; do not label an alarm nuisance where current owner/competent assessment is absent. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-064 Do we hold current statutory inspection and test evidence for this M&E asset, and exactly where are the gaps?

SENSORS AND TELEMETRY

Proposed: Runtime/status may provide contextual asset identity or service history only; telemetry is never used as a substitute for certificates, inspections or competent-person tests. Coverage: The nominated regulated/statutory asset classes and their authoritative asset identities. Timing: Current document/status snapshot with due/expiry dates from authoritative records; no "compliance score" is generated from sensor data. Limits: Telemetry cannot verify inspection scope, test method, defect severity, certificate authenticity or closure of remedial actions. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation. Decision: At the frequency required by the responsible owner and before work/design depending on regulated equipment. Minimum evidence: Authoritative asset list, applicable current inspection/test requirement as determined by the responsible owner, valid records and defect closure evidence. Method: Map each asset to required evidence and dates, distinguish missing/expired/open-defect states and report provenance without interpreting technical compliance beyond recorded outcomes. Uncertainty: The service may not know every applicable legal duty and cannot assess certificate quality beyond the available authoritative record. Safe failure: Return RECORD COMPLETE / GAP / CONFLICT and route any compliance judgement to the duty holder/competent professional.

WHY THIS IS USEFUL

The engineer can identify exactly which statutory evidence is current, missing or conflicted and route gaps to the responsible owner before work or assurance activity.

AUTHORITATIVE SUPPORTING DATA

Owners: Relevant statutory duty holder/responsible person, compliance-record owner, maintenance owner and competent service provider. Precedence: Current statutory register and valid competent-person certificates/test reports, defect/closure evidence and asset identity take absolute precedence; engineering telemetry is context only. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Missing certificate, expired validity, mismatched asset ID, open defect or conflict between register and physical asset is shown explicitly and escalated to the responsible owner. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not certify legal compliance, determine statutory applicability independently of the responsible owner, close defects, or infer inspection outcome from telemetry. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

Maintenance, statutory duties and technical risk

ME-065 Before maintenance starts, is the technical isolation information complete enough for competent-person site verification?

SENSORS AND TELEMETRY

Proposed: Read-only breaker/valve state, pressure/current/temperature and dependent plant status used only as pre-work context. Coverage: The work asset, documented energy/service sources, isolation devices, stored-energy points and downstream dependencies. Timing: Current-state snapshot plus evidence from prior authorised isolations where available; no automatic action or "safe" state is generated. Limits: Status indications do not prove isolation, zero energy, valve tightness or absence of backfeed/stored pressure. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + calculation. Decision: During work preparation, before the competent person begins the formal isolation/proving process. Minimum evidence: Current procedure, verified asset/source identities, all documented energy paths and known temporary/modification information. Method: Trace sources and dependencies, enumerate proving points and flag missing/contradictory records; produce a checklist for competent field verification, not an isolation instruction. Uncertainty: Undocumented changes and local stored energy can remain invisible to records/telemetry. Safe failure: Return INCOMPLETE where any source/path is uncertain and require the formal site proving process.

AUTHORITATIVE SUPPORTING DATA

Owners: Permit/isolation procedure owner, electrical/mechanical duty holder/responsible person, maintenance lead and discipline engineer. Precedence: Current approved isolation procedure/schematic and physical asset labels, then competent-person site identification and required proving/test steps; telemetry is contextual only. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Any unresolved alternate feed, local generation, bypass, cross-connection, stored energy or label conflict makes the information package incomplete. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not issue an isolation plan as authority to act, state equipment is safe, instruct switching/valve operations or replace lockout/tagout, prove-dead, depressurisation or permit procedures. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can enter maintenance with a complete information checklist and avoid discovering undocumented dependencies only after the shutdown begins.

ME-066 Which M&E assets carry the highest technical risk from obsolescence, unavailable spares, unsupported controls or specialist dependency?

SENSORS AND TELEMETRY

Proposed: Runtime, fault frequency, controller/firmware identifiers where legitimately available, installed redundancy state and condition indicators. Coverage: Selected critical plant, drives, controls, meters and protection/control interfaces; do not enumerate sensitive network details beyond engineering need. Timing: Monthly/annual operating summaries plus current support/spares evidence at retrieval time. Limits: A legacy asset can be reliable while a newer unsupported component can be risky; telemetry cannot prove future spare availability or remaining life. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + calculation + forecast + diagnosis + recommendation. Decision: Annual capital planning and after support notices, major failures or system changes. Minimum evidence: Verified asset identity/model, current support/spares status, failure/maintenance history, service criticality and known redundancy/recovery route. Method: Score likelihood-of-support-failure and consequence separately; identify single-source spares, unsupported software/controls and long lead-time components; run sensitivity where cost/lead time is uncertain. Uncertainty: Supportability risk is not the same as current condition and exact failure probability is rarely known. Safe failure: Return prioritized engineering mitigation options and evidence gaps; do not predict failure dates or mandate replacement.

AUTHORITATIVE SUPPORTING DATA

Owners: Asset/capital planning owner, maintenance/procurement owner, discipline engineer, approved supplier/OEM support records and cybersecurity owner where controls are involved. Precedence: Current support/end-of-life/spares information and asset criticality/maintenance history, then competent condition assessment and verified operational telemetry. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Supplier support status is time-sensitive; stale quotations/catalogues or uncertain model/firmware identity are flagged with retrieval date. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not expose sensitive control/network inventory beyond authorised need, guarantee spare availability, predict exact asset life or approve procurement. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can build a defensible replacement/spares strategy around assets whose failure would be difficult to recover from, not just those with the most alarms.

ME-067 Are runtime hours and start counts high enough to change the inspection or service priority for this plant?**SENSORS AND TELEMETRY**

Proposed: Verified run feedback, motor current/power confirmation where useful, start/stop events, speed and load proxy. Coverage: The nominated rotating/refrigeration/mechanical plant with authoritative asset identity. Timing: Event-level starts and daily runtime accumulation over the period since the last verified service; retain counter reset/replacement events. Limits: Commanded run may not equal actual run; controller replacement can reset counters; starts have different severity depending on load/sequence. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + recommendation. Decision: Monthly planning and before anticipated high-use periods. Minimum evidence: Verified asset identity, authoritative last-service completion, applicable usage/calendar criteria and continuous enough runtime/start evidence. Method: Reconcile calendar due status with accumulated run/start exposure, identify assets exceeding usage thresholds and compare start intensity with peer operating periods. Uncertainty: Usage does not directly measure wear and low use may not justify extending time-based tasks. Safe failure: Use a conservative due status where counter continuity or rules are uncertain and refer extensions to the maintenance owner.

AUTHORITATIVE SUPPORTING DATA

Owners: Maintenance strategy/CMMS owner, discipline engineer, OEM/service information owner and asset-information owner. Precedence: Current approved maintenance rule and last competent maintenance completion, then verified runtime/start measurement and OEM guidance; mandatory intervals remain authoritative. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Unknown last-service date, counter continuity or equipment replacement prevents precise usage-since-service calculation. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not extend mandatory maintenance intervals, infer wear state from hours alone, or reset counters/CMMS schedules. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can prioritize inspection on heavily cycled plant and identify counter/data quality problems before using runtime-based maintenance logic.

ME-068 Which temporary bypasses, manual workarounds or control overrides have persisted long enough to become a technical risk?**SENSORS AND TELEMETRY**

Proposed: Read-only override/manual/bypass indicators, command-feedback states, alarm suppressions where visible and affected process/service performance. Coverage: Selected plant and control functions with current authorisation records; safety-related bypasses are treated as high-consequence evidence requiring immediate owner visibility. Timing: Current state plus start time/history where available; calculate persistence and recurrence without clearing anything. Limits: Not every physical bypass is visible to BMS, and some long-lived temporary states can be formally authorised while permanent work is pending. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + calculation + estimate + diagnosis. Decision: Daily/weekly for active temporary states and before maintenance/commissioning. Minimum evidence: Verified active state, authoritative authorization/expiry where applicable, affected service dependencies and current risk/mitigation record. Method: Identify states exceeding their authorised window, map affected controls/alarms/redundancy and distinguish known temporary mitigations from unexplained persistence. Uncertainty: Telemetry cannot judge whether the temporary arrangement is physically safe or legally acceptable. Safe failure: Escalate unresolved/expired states to the responsible owner with evidence; do not clear or regularize them automatically.

AUTHORITATIVE SUPPORTING DATA

Owners: Operational/maintenance owner, BMS/controls owner, duty holder/responsible person for affected safety functions and discipline engineer. Precedence: Current authorised temporary-work/bypass record and permit/change authorization, then physical/controls state and competent-person evidence; telemetry alone cannot establish authorization. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: A bypass with no current owner, expiry, reason or matching field state is an unresolved risk and must be escalated rather than normalized. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not remove bypasses, cancel overrides, reinstate safeties, extend temporary authorizations or declare a temporary arrangement safe. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can stop temporary technical debt from disappearing into normal operation and ensure each persistent workaround has an accountable next action.

ME-069 Which access, thermal, noise, moisture or service constraints could make planned maintenance difficult or unsafe to execute?**SENSORS AND TELEMETRY**

Proposed: Plantroom temperature/humidity, water-presence alarms, equipment noise level only from an authorised engineering survey, door/access status where approved, and plant operating state. Coverage: The plant maintenance envelope and access route; no person tracking or access-log profiling. Timing: Current environmental trends over relevant operating periods plus a competent site survey close to the work date. Limits: Fixed sensors cannot prove clear access, lighting, trip hazards, lifting space, asbestos/other hazards, guard condition or temporary obstruction. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation. Decision: At maintenance planning and shortly before mobilization. Minimum evidence: Current work method/equipment envelope, authoritative access/hazard information and competent site confirmation of physical conditions. Method: Compare required access/lifting/clearances and environmental limits with current evidence; flag constraints that need permit, shutdown, temporary ventilation, lifting or access planning by the relevant owner. Uncertainty: Telemetry cannot establish all site hazards or safe working conditions. Safe failure: Return a constraint list and mandatory site-verification items; never state the work area is safe based on sensors/old drawings.

AUTHORITATIVE SUPPORTING DATA

Owners: Maintenance/work planner, Estates space/access owner, relevant safety/hazard record owners, discipline engineer and competent site surveyor. Precedence: Current site access/hazard information and competent pre-work inspection, then manufacturer maintenance-clearance requirements and current plant/environment data; historical plans are context only. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Any changed access arrangement, closure, temporary works or unknown hazard status supersedes old spatial information. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not issue safe-to-work clearance, infer current hazards/closures from the May 2021 plan, track workers, or replace pre-work inspection and permit controls. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can plan access, lifting, shutdown and environmental controls before a contractor arrives and avoid failed maintenance visits.

ME-070 Which M&E technical risks deserve attention first when consequence, evidence quality and recoverability are considered together?**SENSORS AND TELEMETRY**

Proposed: Verified alarms/faults, runtime/condition indicators, energy anomalies, resilience states and service-performance trends relevant to each risk candidate. Coverage: Defined M&E asset/system portfolio with owner-approved criticality boundaries; no person or research-content risk inference. Timing: Use current state plus months of history where available; summarize frequency, duration and evidence completeness separately. Limits: Well-instrumented assets can appear riskier because more is observed; missing telemetry must not be interpreted as good condition. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis + recommendation. Decision: Monthly/quarterly engineering planning and after significant incidents/defects. Minimum evidence: Current criticality/consequence criteria, authoritative open defects/maintenance status and enough evidence to estimate recurrence/condition for candidate risks. Method: Separate consequence, likelihood evidence, detectability, redundancy/recovery and evidence confidence; present Pareto-style priorities plus sensitivity to uncertain scores and keep mandatory actions outside discretionary ranking. Uncertainty: Risk scores are decision aids, not quantified failure probabilities, and data-rich assets can bias comparisons. Safe failure: Escalate mandatory/high-consequence unknowns; label low-confidence rankings and request the evidence that could change priority.

WHY THIS IS USEFUL

The engineer can prioritise investigation and mitigation by technical consequence and recoverability rather than by raw alarm count or anecdote.

AUTHORITATIVE SUPPORTING DATA

Owners: Discipline engineers, asset/maintenance owner, relevant duty holders/responsible persons, resilience/service owners and risk register owner. Precedence: Mandatory/statutory safety actions and current owner-approved criticality/risk records take precedence, then competent condition/failure evidence; telemetry supplies current operating evidence only. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Unknown criticality, open statutory defect, unverified asset identity or missing condition evidence is shown and conservatively escalated rather than assigned a low score. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not downgrade statutory/safety actions because telemetry looks normal, claim quantified safety risk, or automatically set maintenance/capital priorities. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-071 What do measured heating demand, temperature levels and electrical constraints imply for early heat-pump retrofit feasibility?**SENSORS AND TELEMETRY**

Proposed: Verified heat meter or flow-and-temperature data, plant supply/return temperatures, heating runtime, outdoor temperature and building electrical demand at the relevant supply boundary. Coverage: Heating plant/distribution system plus the upstream electrical supply/boards likely to serve heat-pump equipment; no room-use inference from heat demand. Timing: At least one representative heating season where possible, retaining cold-weather peak intervals and daily load/temperature relationships. Limits: Existing boiler input, valve position or gas consumption alone may not equal delivered heat; a mild winter may miss design peak and low-temperature terminal performance is not proven by aggregate data. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + recommendation. Decision: At option development before detailed heat-loss, emitter, hydraulic and electrical design. Minimum evidence: Credible delivered-heat profile, relevant cold-weather data, current operating temperatures, plant/distribution design information and verified electrical baseline/constraints. Method: Estimate weather-related heat demand and peak range, test lower flow-temperature operating periods where evidence exists, and translate candidate heat-pump input into electrical peak scenarios with sensitivity for COP/defrost/backup. Uncertainty: Observed peaks may not equal design heat loss; terminal capacity, acoustics, refrigerant, space, network and protection constraints remain separate design tasks. Safe failure: Return a feasibility range and critical surveys/calculations, not equipment size or approval.

WHY THIS IS USEFUL

The engineer can decide whether heat-pump design is plausible, which temperature/capacity constraints dominate and what detailed surveys should be commissioned next.

AUTHORITATIVE SUPPORTING DATA

Owners: Mechanical/electrical design engineers, energy owner, current heat-source/distribution record owners and project brief owner. Precedence: Current heat-load/design-temperature criteria, current system/terminal data and electrical capacity/protection records, then calibrated heat/electrical measurements; scenario assumptions remain analytical. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Changed use, incomplete heat metering, unknown terminal duties or missing electrical design data prevents a definitive feasibility conclusion. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not size or approve heat pumps, guarantee COP/savings/carbon reduction, or claim electrical/terminal capacity from telemetry alone. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-072 How much practical headroom does the existing ventilation system appear to have for higher airflow or a revised ventilation strategy?**SENSORS AND TELEMETRY**

Proposed: Measured AHU/branch airflow where valid, fan speed/power, static pressure, damper positions, filter DP and terminal airflow for critical branches. Coverage: The affected AHU, main distribution and critical terminals/zones with current verified mappings. Timing: Representative high-demand occupied periods and controlled test points if authorised; use stable 5-15 min windows across fan-speed range. Limits: Fan speed percentage is not airflow headroom; duct noise, diffuser throw, pressure limits, motor/VSD capacity and terminal limits may constrain before fan maximum. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + recommendation. Decision: During change-of-use, ventilation retrofit or increased-airflow option assessment. Minimum evidence: Current required/proposed airflow, verified existing airflow/pressure, fan/motor/VSD selection data and evidence for critical terminal/branch constraints. Method: Locate current operating points, compare with documented fan/system limits, identify saturated dampers/pressure constraints and test scenario airflow ranges with explicit uncertainty. Uncertainty: System curve and acoustic/distribution performance may change nonlinearly; increased fan speed may not deliver proportional zone airflow. Safe failure: Return an indicative headroom range and required fan/duct/terminal measurements/design checks.

WHY THIS IS USEFUL

The engineer can tell whether a ventilation option deserves detailed design or is likely constrained by fan, duct, terminal or control limitations.

AUTHORITATIVE SUPPORTING DATA

Owners: Mechanical design engineer, ventilation commissioning/balancing owner, current system drawings/selection data owner and BMS owner. Precedence: Current fan/AHU selection and duct/terminal design data, latest balancing/commissioning evidence and current design airflow requirements, then verified telemetry. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Altered ductwork/terminals, stale balancing, unknown fan curve or unverified airflow station prevents quantified headroom. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not approve higher airflow, change fan/VSD/pressure settings, certify ventilation/noise performance or infer duct capacity from speed alone. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-073 What does the current electrical demand profile imply for PV and/or battery-storage opportunity and constraints?**SENSORS AND TELEMETRY**

Proposed: Building-import/export power, verified major load submeters where useful, existing generation meter if present and power-quality/voltage only where appropriately measured. Coverage: The electrical supply boundary and candidate generation/storage connection area; solar-resource data may come from an authoritative external/site source rather than inferred sensors. Timing: 15 min or finer annual demand where available, preserving seasonal peaks, minimum daytime demand and any export periods. Limits: Historical load does not establish future electrification demand; rooftop area/structure/shading, fault level, protection and network-connection limits are not inferable from demand telemetry. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + calculation + estimate + diagnosis + recommendation. Decision: At option appraisal before detailed structural/electrical design and connection application. Minimum evidence: Verified annual demand profile, current supply/export constraints, credible solar/storage performance assumptions and candidate system boundaries. Method: Simulate interval self-consumption, import/export, peak shaving and battery cycling across size ranges; report sensitivity to future loads, degradation, control objective and connection limit. Uncertainty: Economic/carbon outcome depends on future tariffs/factors and technical feasibility still requires roof, protection, fault-level and fire/safety design. Safe failure: Return technically plausible ranges and hard evidence gaps; do not name a final system size.

WHY THIS IS USEFUL

The engineer can identify whether PV, storage or both align with the actual load shape and which network/roof/design constraints need early resolution.

AUTHORITATIVE SUPPORTING DATA

Owners: Electrical design engineer, energy owner, DNO/connection-information owner where applicable, Estates roof/asset owner and project brief owner. Precedence: Current supply/connection/protection data and approved project constraints, current roof/structural/site survey and authoritative solar assumptions, then validated demand/generation measurements. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Provisional tariffs, carbon factors, future loads, roof condition or connection limits are scenario inputs with retrieval/effective dates, not fixed facts. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not approve PV/battery design, guarantee payback/savings, submit a network application, certify roof/fire/protection suitability or control batteries. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-074 What might a lighting retrofit save while still meeting current lighting design and control requirements?**SENSORS AND TELEMETRY**

Proposed: Lighting circuit/submeter power where available, BMS/lighting-control run state, approved aggregate occupancy schedule only as operational context, and competent illuminance survey results. Coverage: Selected lighting zones/circuits with current space/use and luminaire/control mapping; no person tracking. Timing: 15 min power over representative weeks/seasons plus exact times/conditions of current illuminance surveys; summarize runtime by authorised service window. Limits: Circuit power may include non-lighting loads; dimming percentage is not luminaire power; historical room use and old illuminance readings may not match current requirements. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + recommendation. Decision: At retrofit option appraisal before detailed lighting design. Minimum evidence: Current lighting requirement, verified existing inventory/control strategy, credible circuit energy/runtime baseline and representative illuminance evidence. Method: Separate power reduction, runtime/control reduction and maintenance effects; model candidate wattage/control changes while retaining required illuminance/uniformity/glare/emergency interfaces as design constraints to be checked. Uncertainty: Telemetry cannot prove photometric compliance and future use can change runtime. Safe failure: Report energy range and required detailed lighting calculations/survey; no final luminaire design is selected.

WHY THIS IS USEFUL

The engineer can distinguish savings from efficient luminaires versus better controls and know what photometric/design work remains before procurement.

AUTHORITATIVE SUPPORTING DATA

Owners: Electrical/lighting design engineer, current space/use owner, lighting-control owner and energy/metering owner. Precedence: Current lighting criteria/brief and competent illuminance survey, current luminaire/control/circuit records, then validated energy/runtime telemetry and manufacturer photometric data for scenarios. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: If current use/criteria, circuit boundary or existing luminaire inventory is uncertain, savings/service comparison remains indicative. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not certify lighting compliance, emergency-lighting performance, approve luminaires/controls or infer occupancy from lighting state. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-075 Which constant-speed or poorly modulated fans and pumps are the strongest VSD or controls retrofit candidates?**SENSORS AND TELEMETRY**

Proposed: Motor electrical power/current, speed/frequency, flow or pressure where valid, valve/damper positions and runtime. Coverage: Selected motors/fans/pumps with authoritative motor/drive and system mapping. Timing: 10-60 s operating points across representative load periods; summarize fraction of runtime at throttled/low-demand conditions and starts. Limits: Throttle position is not wasted energy by itself; process minimums, motor type, resonance, cooling, harmonics and control stability can limit VSD suitability. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + calculation + estimate + forecast + recommendation. Decision: During energy/capital option development before detailed VSD/control design. Minimum evidence: Verified runtime/power and credible flow/pressure/demand evidence, current process constraints and motor/drive characteristics. Method: Identify sustained throttled or part-load operation, estimate speed-reduction range only within valid fan/pump affinity conditions, model energy range and flag control/process/electrical checks. Uncertainty: Affinity laws fail across some system changes and cannot predict final control stability or equipment efficiency accurately. Safe failure: Return candidate ranking and required detailed design/test evidence, not a VSD approval.

WHY THIS IS USEFUL

The engineer can focus detailed VSD studies on equipment with measured part-load opportunity instead of converting assets that already operate near required duty.

AUTHORITATIVE SUPPORTING DATA

Owners: Mechanical/electrical design engineers, motor/drive asset owner, commissioning owner and energy/BMS owner. Precedence: Current motor/fan/pump selection, process minimum/maximum requirements and electrical/protection information, then verified measured duty; retrofit vendor estimates are scenario inputs only. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Unknown motor compatibility, driven-equipment curve, process minimum or electrical harmonic/protection constraint prevents a technical recommendation. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not specify drive settings, approve VSD installation, guarantee energy savings or ignore motor/protection/harmonic/process constraints. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-076 How much performance or energy opportunity may exist from controls optimisation before major plant replacement?**SENSORS AND TELEMETRY**

Proposed: Schedules, setpoints, modes, valve/damper saturation, fan/pump speed, plant enable, energy/power and service variables such as temperatures/pressures. Coverage: Selected plant systems with current approved service requirements and verified control-point mappings. Timing: 1-15 min data over representative seasons; identify repeatable control inefficiencies and retain event detail for sequence issues. Limits: Observed inefficiency may reflect sensor/valve/plant faults rather than poor sequence; optimization potential can disappear after maintenance or change of use. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis. Decision: Before plant replacement business cases and during recommissioning planning. Minimum evidence: Current control intent, validated service/energy measurements and enough data to show persistent schedule/reset/overlap/saturation patterns under representative conditions. Method: Quantify avoidable runtime/overlap or model transparent reset/schedule scenarios while checking service constraints; rank low-cost tests by expected value and uncertainty. Uncertainty: Counterfactual energy savings are model-dependent and control changes can shift loads elsewhere or degrade service. Safe failure: Return a testable optimization shortlist with savings ranges; require controlled implementation and proving before any permanent change.

WHY THIS IS USEFUL

The engineer can determine whether recommissioning/controls work should precede capital replacement and which changes are worth controlled testing.

AUTHORITATIVE SUPPORTING DATA

Owners: Discipline engineer, BMS/controls owner, current service-requirement owner and energy owner. Precedence: Current approved sequence/setpoints/service criteria and competent commissioning/fault evidence, then verified telemetry; proposed optimization logic is a scenario until designed/approved/tested. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Unknown control intent, unverified sensors or active temporary overrides prevent a credible optimization estimate. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not change setpoints/sequences, guarantee savings, bypass protective modes or treat optimization as a substitute for defective plant maintenance. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

Retrofit, capacity, lifecycle, decarbonisation and scenarios

ME-077 Which M&E assets should be prioritised for lifecycle replacement when condition, criticality, energy, supportability and intervention risk are combined?

SENSORS AND TELEMETRY

Proposed: Runtime, verified faults/alarms, energy intensity and asset-specific condition indicators where technically valid. Coverage: The capital-planning asset portfolio with authoritative asset identities and criticality; missing telemetry is recorded as uncertainty, not good condition. Timing: Monthly/seasonal summaries with latest competent condition evidence and current support/spares status. Limits: Remaining life is rarely directly measurable; instrumented assets can appear worse because more is known and energy performance may be dominated by system control rather than the asset. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis + recommendation. Decision: Annual capital planning and after major failures/support withdrawals. Minimum evidence: Current asset portfolio, agreed criteria/weights or scenario ranges, mandatory actions separated out, and evidence-quality score for each material input. Method: Compare consequence, condition, reliability, energy, obsolescence, maintainability, intervention risk and cost; run sensitivity to weights/uncertain life/cost and identify robust versus fragile rankings. Uncertainty: Scores are not failure probabilities and exact remaining life/payback is uncertain. Safe failure: Flag assets whose ranking is driven by missing/weak evidence and specify condition survey or support check before commitment.

WHY THIS IS USEFUL

The engineer can build a transparent replacement shortlist, defend why assets rank highly and see where better evidence could materially change investment order.

AUTHORITATIVE SUPPORTING DATA

Owners: Asset/capital-planning owner, maintenance owner, energy owner, discipline engineers and procurement/support information owners. Precedence: Mandatory/statutory defects and owner-approved criticality first, then current condition/failure history, support/spares status and approved strategy/cost information; verified telemetry is supporting operational evidence. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Supplier lead times/costs and condition evidence are time-sensitive; unknown asset age/model or conflicting failure records reduce ranking confidence. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not predict exact remaining life, approve capital expenditure, downgrade mandatory defects or present the ranking as a professional asset-management decision by itself. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-078 If a space changes use, which M&E capacity and service constraints should be verified first before detailed design?

SENSORS AND TELEMETRY

Proposed: Existing aggregate zone electrical demand, verified ventilation airflow/CO2 indicators, heating/cooling duty indicators and water demand at appropriate engineering boundaries. Coverage: The proposed space plus serving electrical, HVAC and water branches with current verified mappings; no inference of future occupancy from historical sensor patterns. Timing: Representative existing-use weeks/seasons and peak intervals; scenario inputs use the proposed design brief rather than extrapolated current behavior. Limits: Existing measured demand does not define future design load; ventilation, heat gains, water, acoustics, resilience and specialist services depend on the proposed use/occupancy/equipment brief. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + recommendation. Decision: At project feasibility before discipline detailed design. Minimum evidence: Approved initial use/equipment brief, verified serving-system mappings and current evidence for electrical/HVAC/water capacities and service constraints. Method: Create discipline constraint matrix: electrical demand/quality/resilience, ventilation/thermal load, controls, water/drainage and maintainability; compare existing measured operation only as baseline and list design calculations/surveys required. Uncertainty: Current operational headroom cannot establish future design adequacy and some constraints are spatial/statutory rather than measurable. Safe failure: Return constraints and required design evidence; do not declare the space suitable.

WHY THIS IS USEFUL

The engineer can identify fatal or expensive service constraints early and commission the right load, airflow, drainage and electrical studies before design develops.

AUTHORITATIVE SUPPORTING DATA

Owners: Project brief owner, current space-information owner, electrical/mechanical/public-health design engineers and Estates service-record owners. Precedence: Current approved proposed-use/load brief and current authoritative room/service mappings, then current design/capacity records and competent surveys/tests; the May 2021 floor plan is historical only. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: If the proposed use/equipment/occupancy brief or current service mapping is provisional, only a gap/constraint screen is possible. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not approve change of use, infer current or future occupancy from telemetry, certify capacity/safety or rely on the 2021 room label as current evidence. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

ME-079 What is the defensible operational energy and carbon baseline by major M&E end use, with allocation uncertainty visible?**SENSORS AND TELEMETRY**

Proposed: Validated electricity, gas/heat and water-energy-related meters plus mapped M&E submeters and plant runtime used for limited allocation where physically justified. Coverage: Whole-building energy boundary and major M&E end-use groups with verified meter/system mapping; no person/device/research-activity disaggregation. Timing: At least 12 representative months where available, using native intervals for reconciliation and monthly/seasonal totals for baseline; carbon factors carry explicit effective dates. Limits: Submeter gaps, shared services, embedded generation and changing use create residual energy; runtime-based allocation can be highly uncertain and must be labelled modeled. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + estimate + forecast. Decision: Annual/seasonal decarbonisation baselining and before retrofit comparisons. Minimum evidence: Validated whole-building energy data, verified submeter map, current factor set and an explicit allocation method for each unmetered residual. Method: Reconcile metered end uses to main totals, keep residual separate, use modeled allocation only where assumptions are declared, calculate carbon by effective-period factors and provide uncertainty/range by end use. Uncertainty: Operational carbon is not whole-life carbon and end-use allocation may be less certain than building total. Safe failure: Do not force 100 percent allocation; label residual/unverified end uses and request metering only where it changes decisions.

AUTHORITATIVE SUPPORTING DATA

Owners: Energy/metering owner, discipline engineers, utility/billing record owner and current carbon-factor source owner used by the organisation. Precedence: Validated utility/main-meter data and current meter hierarchy first, then verified submeters; approved organisational carbon factors with effective dates; modeled end-use allocations are lowest precedence. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Meter replacements, incomplete periods, changing fuel factors or unresolved meter boundaries are segmented and reported rather than smoothed away. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not present modeled allocation as measured energy, claim whole-life carbon, infer individual activity, guarantee future carbon savings or choose organisational carbon factors without the authoritative owner. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can target the largest evidenced M&E energy/carbon contributors and avoid decarbonisation priorities built on false precision.

ME-080 Which retrofit scenarios remain technically robust when capacity, service performance, lifecycle, implementation constraints and uncertainty are tested together?**SENSORS AND TELEMETRY**

Proposed: A common verified baseline of energy, peak demand, plant duty/condition and service-performance variables relevant to every candidate option. Coverage: The same building/system boundary for all scenarios, with option-specific extra measurements clearly identified rather than hidden by a common score. Timing: Historical periods covering material seasonal/peak behavior; all scenario inputs retain effective/observation dates and use common baseline windows where possible. Limits: Different options need different evidence granularity; future weather/use/prices, equipment performance and project interfaces can reverse rankings. Quality gate: verify point/asset/location, units, calibration or field check, clock alignment, cadence/aggregation, uptime/gaps/stale repeats, drift/outliers, provenance, permission and current health; otherwise NOT VERIFIED. Asset/zone outputs only; no person tracking or activity/content inference.

ANALYSIS REQUIRED

Operations: lookup + observation + comparison + calculation + diagnosis + recommendation. Decision: At option appraisal and design gateways before a preferred scheme is approved. Minimum evidence: Common verified baseline, explicit option assumptions, current hard constraints, comparable lifecycle boundary and evidence-quality rating for every major input. Method: Reject scenarios that fail hard constraints; compare remaining options across capacity margin, service effect, energy/carbon, lifecycle, maintainability and implementation risk using ranges; run sensitivity to uncertain inputs and show robust versus assumption-sensitive rankings. Uncertainty: Future performance, prices, carbon factors, use and interfaces are uncertain; a single weighted score can hide technical trade-offs. Safe failure: Return a transparent option matrix, evidence gaps and sensitivity, leaving final design/investment choice to the project authority and competent designers.

AUTHORITATIVE SUPPORTING DATA

Owners: Project sponsor/design team, Estates discipline owners, energy/capital owners, current asset/space owners and relevant duty holders for constraints. Precedence: Current approved project requirements and hard constraints, verified capacity/service/lifecycle evidence and current cost/carbon assumptions with dates; telemetry/model scenarios follow and never override mandatory requirements. Evidence times/provenance: keep issue/effective, test/observation, telemetry and retrieval times separate, with permission, asset/point ID and lineage. Staleness/conflict: Any option depending on unverified capacity, stale cost/support data or unresolved service constraint carries an explicit feasibility/evidence penalty rather than optimistic assumptions. Conflicts stay visible and go to the current authoritative owner or competent professional; analytics never silently resolves them.

ANSWER BOUNDARY

Do not approve a preferred design/investment, guarantee savings/payback/carbon outcomes, hide infeasible assumptions or substitute scenario analytics for discipline design and professional sign-off. Service boundary: no professional engineering sign-off, design approval, compliance/safety certification, permit or work/isolation authorisation, switching, plant operation, resets, setpoint changes or BMS/control-logic changes. Telemetry never substitutes for required inspection, testing, commissioning or competent-person judgement.

WHY THIS IS USEFUL

The engineer can identify which options survive hard constraints and uncertainty, which fail, and which missing evidence would most improve the project decision.

From evidence to a safe engineering next step

1

Bound the question

Name the asset, circuit, plant, zone, service and decision interval. Do not silently widen the spatial or operational boundary.

2

Check authority

Resolve the current owner, controlled record, permission and applicable effective/test dates before telemetry is used to support a claim.

3

Check measurement fitness

Verify exact point mapping, modality, units, calibration/field check, clock, cadence, aggregation, uptime, gaps, drift and provenance.

4

Use the right operation

Label lookup, observation, comparison, calculation, estimate, forecast, diagnosis and recommendation separately. Keep assumptions visible.

5

Fail safely

If minimum evidence is missing or conflicted, return NOT VERIFIED, explain why, and state the smallest competent survey, inspection, test or record needed next.

NEVER PROMOTE TELEMETRY BEYOND ITS ROLE

- Breaker or status feedback is not proof of electrical topology or safe isolation.
- Fan status, CO2 or damper position is not direct proof of delivered ventilation.
- Valve position, temperature or pump state is not proof of hydraulic flow unless the required measurement exists.
- BMS trends are not statutory, hygiene, commissioning or compliance certification.
- Correlation, temporal order and model fit do not by themselves establish causation.

EXPECTED ENGINEERING OUTPUT

- A bounded statement with source and time context.
- An evidence state: SUPPORTED, INDICATIVE, CONFLICTED or NOT VERIFIED.
- Uncertainty and unmeasured dependencies left visible.
- A targeted next action: record correction, survey, portable measurement, test, maintenance, recommissioning, design study or owner escalation.
- No automatic control action or safety/compliance claim.

Use data only where its authority, meaning and limits are known

Current authority first

Controlled engineering records, duty holders, responsible persons and competent professionals govern present engineering facts. Telemetry corroborates; it does not silently overrule them.

Provenance stays attached

Carry source owner, asset/point ID, units/scaling, mapping confidence, calibration/field check, observed/test/effective/retrieval times, permission and quality state into the answer.

Historical floor plan only

The May 2021 floor plan may orient historical spatial context. It cannot establish current room use, access, service routes, hazards, closures or safe work areas.

Conflicts are not averaged away

When controlled records, competent observations/tests or verified telemetry disagree, show the disagreement and refer it to the authoritative owner or competent professional.

Least-privilege access

Expose only evidence the user is authorised to see. Restrict critical infrastructure detail where necessary. Keep outputs at asset, circuit, plant or zone level.

Safe failure is an answer

If minimum evidence is stale, incomplete, unpermissioned, mis-mapped, unhealthy or unverified, state NOT VERIFIED and identify the next safe evidence step rather than filling the gap by inference.

Privacy by construction

Do not identify or track people, infer individual behaviour/productivity, inspect research content, code, speech or network content, or use personal-device identity as engineering evidence.

Handover remains human-led

Recommendations support engineering judgement. They do not approve designs, issue permits, certify compliance/safety, operate or isolate systems, or substitute for competent inspection, testing and commissioning.

CATALOGUE COMPLETENESS

ME-001-010

ME-011-020

ME-021-030

ME-031-040

ME-041-050

ME-051-060

ME-061-070

ME-071-080